CHAPTER XXVI

ON 'CONSCIOUSNESS' AND CONSCIOUSNESS OF ABSTRACTING

> But a felt 'contrary' is consciousness in germ. . . . Consciousness requires more than the mere entertainment of theory. It is the feeling of the contrast of theory, as *mere* theory, with fact, as *mere* fact. This contrast holds whether or no the theory be correct. (578) A. N. WHITEHEAD

A language, to be most useful, should be similar in its structure to the structure of the events which it is supposed to represent. The language of 'abstractions of different orders' appears to be satisfactory in point of structure. It is a *non-el* language, since it does not discriminate between 'senses' and 'mind',. It is a functional language, since it describes, by implication, what is going on in the nervous system when it reacts to stimuli. It is a language which can be made as flexible and as sharp as desired, thus making it possible to establish sharp verbal differences, of both horizontal and vertical type, between the terms 'man' and 'animal'.

The last semantic characteristic of potential sharpness is extremely important for a theory of sanity. Evidence of 1933 leads us to conclude that, under the influence of external stimuli, the most primitive and simplest forms of life were moulded, transformed, and influenced in the process of survival, and, therefore, of adjustment. In this way, more and more complex structures evolved. It should be emphasized that organisms represent *functional* units, and that an additive change in structure does not necessarily involve a *simply additive* change in function. By physico-chemical, structural, colloidal necessity the organism works as-a-whole. Being a relative whole, any additive structural factor becomes a reactive and functional factor which influences the working of the whole. This is, perhaps, best illustrated by the boy who was born without a cortex, but with no other obvious defects. He was incomparably more helpless and unadjusted than animals who have no cortex, or even no nervous system at all. Although we could speak in *additive* terms of the difference between this boy and a normal boy, as one having no cortex and the other 'plus a cortex', yet the functioning was so different as not to be expressible in a 'plus' language.

Similar remarks could be generalized to all life. We must be very careful in building sharp distinctions, since the anatomical differences alone are unreliable. If we want to have more reliable differences, we should look for *functional* differences.

We have already discovered *functional* differences that are expressed by the horizontal and by the vertical differences between the abstracting capacities of Smith and Fido. The analysis of these differences is the subject of the present chapter.

'Thought' represents a reaction of the organism-as-a-whole, produced by the working of the whole, and influencing the whole. From our daily experience, we are familiar with what we usually denote as being 'conscious'; in other words, we are aware of something, be it an object, a process, an action, a 'feeling', or an 'idea'. A reaction that is very habitual and semi-automatic is not necessarily 'conscious'. The term 'consciousness', taken separately, is not a complete symbol; it lacks content, and one of the characteristics of 'consciousness' is to have some content. Usually, the term 'consciousness' is taken as undefined and *undefinable,* because of its immediate character for every one of us. Such a situation is not desirable, as it is always semantically useful to try to define a complex term by simpler terms. We may limit the general and undefined term 'consciousness' and make it a definite symbol by the deliberate ascribing of some content to this term. For this 'consciousness of something' I take 'consciousness of abstracting' as fundamental. Perhaps the only type of meanings the term 'consciousness' has is covered by the functional term 'consciousness of abstracting', which represents a general process going on in our nervous system. Even if this is not the only type of meanings, the term 'consciousness of abstracting' appears to be of such crucial semantic importance that its introduction is necessary.

The term 'consciousness', because of its hitherto undefined and traditionally *undefinable* character, did not allow us further analysis. Neither did we have any *workable,* educational, semantic means to handle the vast field of psycho-logical processes which this incomplete symbol indicated. If we now select the term 'consciousness of abstracting' as fundamental, we not only make the last symbol complete by assigning functional content to it, but we also find means to define it more specifically in *simpler terms.* Through understanding of the processes we gain educational means of handling and influencing a large group of semantic psycho-logical reactions.

Let us analyse this new term by aid of the diagram called the Structural Differential referred to in the previous chapter. Here the object (O_h) represents a nervous abstraction of a low order. In this abstracting, some characteristics of the event were missed or not abstracted; these are indicated by the not connected lines (B′). When we abstracted from our object further, by coining a definition or ascribing 'meanings'

to the label (L), again we did not abstract 'all' the characteristics of the object into the definition; but some characteristics were left out, as indicated by the lines (B″). In other words, the number of characteristics which we ascribe to the label, by some process of 'knowing', or 'wanting', or 'needing', or 'interest'., does *not* cover the number of characteristics the object has. The 'object' has more characteristics than we can include in the explicit or implicit definition of the label for the 'object'. Besides, the definition (implicit or explicit) of the 'object' *is not* the object itself, which always holds many surprises for us. The latter has the *'individuality of the object'*, as we may call it. Every one who uses a car, or a gun, or a typewriter, or who has had a number of wives, or husbands, or children, knows that well. In spite of the fact that these objects are, to a large extent, standardized, every individual object has individual peculiarities. With modern methods of physical, chemical, and astronomical investigation, scientists find that even their special materials and equipments have also peculiar individualities which must be taken into account in the more refined researches.

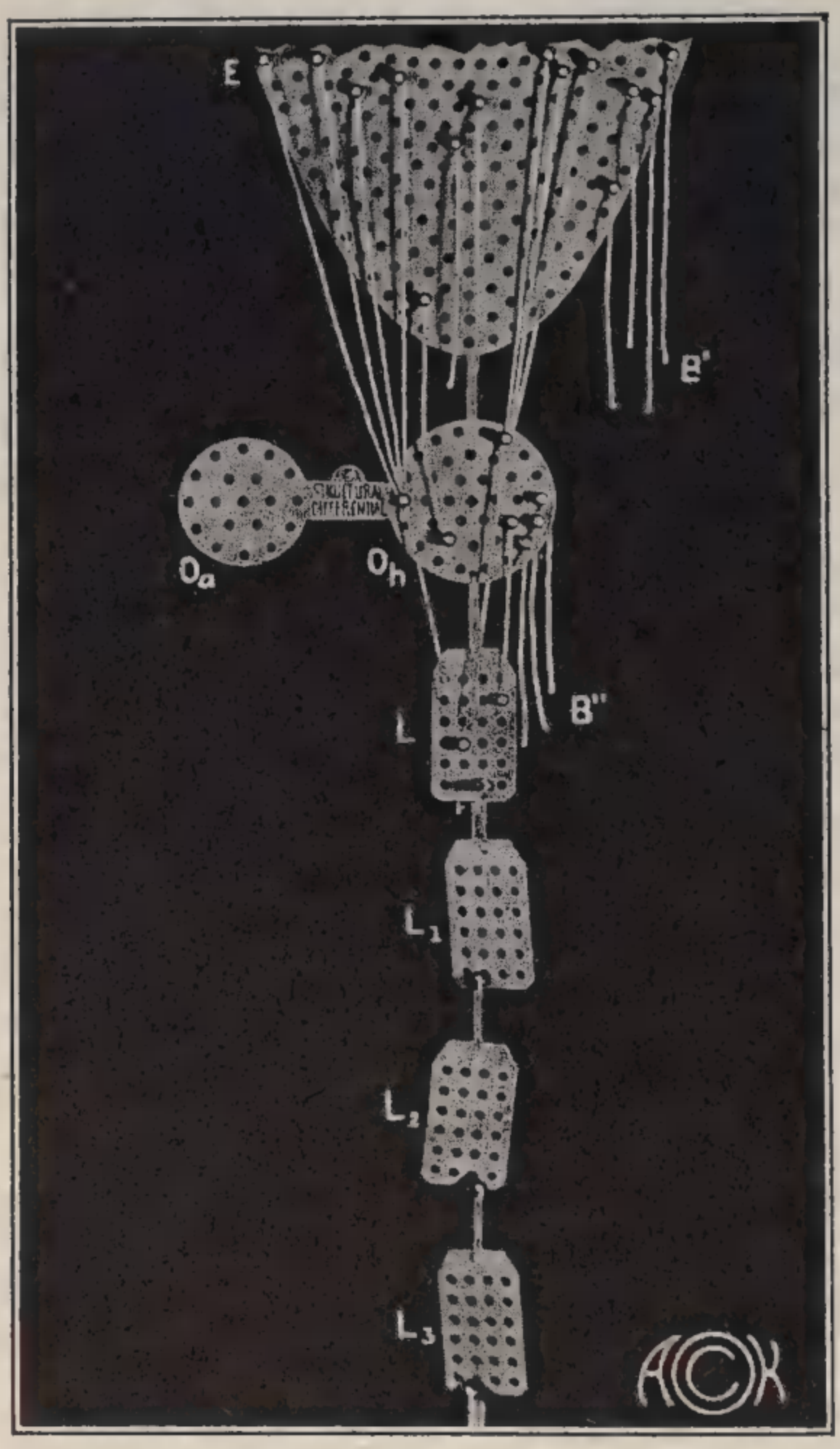


FIG. 1
THE STRUCTURAL DIFFERENTIAL

If we take any ordinary object and expect to find such and such characteristics, ascribed to the objects by *definition,* we may be disappointed. As a rule, we find or can find, if our analysis is subtle enough, these peculiar individualities. The reader can easily convince himself

by looking over a box of matches, and by noticing the peculiar individuality of each match. But since, *by definition*, we expect that when we strike a match it should ignite, we may disregard all other characteristics as irrelevant for our purpose. A similar process is at work in other phases of life. We often live, feel happy or unhappy, *by what actually amounts to a definition*, and not by the empirical, individual facts less coloured by semantic factors. When $Smith_1$ marries $Smith_2$, they mostly do so *by a kind of definition*. They have certain notions as to what 'man', 'woman', and 'marriage' 'are' *by definition*. They actually go through the performance and find that the $Smith_1$ and his wife, $Smith_2$, have unexpected likes, dislikes, and particularities—in general, characteristic and semantic reactions *not included* in their definition of the terms 'man', 'woman', 'husband', 'wife', or 'marriage'. Characteristics 'left out' in the definitions make their appearance. 'Disappointments' accumulate, and a more or less unhappy life begins.

The above analysis applies to all phases of human life, and appears entirely general because of the structure of 'human knowledge'. Characteristics are discovered when it is *too late*. The *not knowing* or the *forgetting* of the relations explained above does the semantic havoc. On verbal, 'definitional', or doctrinal semantic grounds, we expect something else than what the experiences of life give us. The non-fulfillment of expectation produces a serious affective and semantic shock. If such shocks are repeated again and again, they disorganize the normal working of the nervous system, and often lead to pathological states. An indefinitely large number of experimental facts fully supports the above conclusions. Many of them have been supplied during the World War. Curiously enough, when the soldier *did* expect horrors, and later experienced them, he seldom became deranged 'mentally'. If he did not fully expect them, and yet had to experience them, he often broke down nervously.

The attack of hay fever at the sight of *paper roses*, referred to already, gives a similar semantic example. The attack followed from the semantic *'definition'* of 'roses', of 'hay fever', and from the situation as-a-whole, and was not due to *inspection* of the objective 'roses', or to the physico-chemical action of the 'roses'. If the patient had been blindfolded when the paper 'roses' were brought into his presence, no attack would have occurred.

We are now ready to define 'consciousness of abstracting' in *simpler terms;* namely, in terms of 'memory'. The term 'memory' is structurally a physico-chemical term. It implies that the events are interconnected,

that everything in this world influences everything else, and that happenings leave some traces somewhere.

A similar analysis can be carried on in connection with the object and the event. Briefly, the object represents structurally an abstraction of some order, does not, and cannot, include all the characteristics of the event; and so, again, we have some characteristics *left out* as indicated by the lines (B′).

Here we have the possibility of making a series of most general, and yet entirely true, *negative* statements of great semantic importance; that the label *is not* the object, and that the object *is not* the event,. For the number of *m.o* characteristics which we ascribe to the label by *definition* does not cover all the characteristics we recognize in the object; and the number of characteristics which we perceive in the object is also not equal to the infinite numbers of characteristics the event has. The differences are still more profound. Not only do the numbers of *m.o* characteristics differ, but also the *character* of these abstractions differs from level to level of the successive abstractions.

We can now define 'consciousness of abstracting' as '*awareness* that in our process of abstracting we have *left out* characteristics'. Or, consciousness of abstracting can be defined as '*remembering* the *"is not"*, and that some characteristics have been *left out*'. It should be noticed that in this formulation, with the aid of the Structural Differential, we have succeeded in translating a *negative* process of forgetting into a *positive* process of *remembering* the denial of identity and that characteristics are left out. Such a positive formulation makes the whole system workable and available for the semantic training and education.

The use of the Structural Differential becomes a necessity for any one who wants to receive full semantic benefit from the present work. A book is, by necessity, *verbal*. Whatever any author can say is verbal, and nothing whatsoever can be *said* which is *not verbal*. It seems entirely obvious that in life we deal with an enormous number of things and situations, 'feelings'., which are *not verbal*. These belong to the 'objective level'. The crucial difficulty is found in the fact that whatever can be said *is not* and *cannot* be on the objective level, but belongs *only* to the verbal levels. This difference, being *inexpressible* by words, cannot be expressed by words. We must have *other means* to indicate this difference. We must show with our hand, by pointing our finger to the object, and by being silent outwardly as well as inwardly, which silence we may indicate by closing our lips with the other hand. The verbal denial of the 'is' of identity covers this point also when shown on the Differential. If we burst into speech based on the 'is' of identity, as we

usually do, we find ourselves obviously on the verbal levels indicated by the labels L, L_1, L_2, . . . L_n, but never on the objective level (O_h). On this last level, we can look, handle., but *must be silent.* The reason that we nearly all identify the two levels is that it is impossible to train an individual in this semantic difference by *verbal means alone,* as all verbal means belong to the levels of labels and never to the objective unspeakable levels. With a visual and tactile *actual object* and labels on the Structural Differential, to point our finger at, handle., we now have simple means to convey the tremendously important semantic difference and train in *non-identity.*

We should notice that the consciousness of abstracting, or the remembering that we abstract in different orders with omission of characteristics, depends on the denial of the 'is' of identity and is connected with limitations or 'non-allness', so characteristic of the new non-systems.

The consciousness of abstracting eliminates *automatically* identification or 'confusion of the orders of abstractions', both applying to the semantic confusion on all levels. If we are *not* conscious of abstracting, we are bound to identify or confuse the object with its finite number of characteristics, with the event, with its infinite numbers of *different* characteristics. Confusion of these levels may misguide us into semantic situations ending in unpleasant shocks. If we acquire the consciousness of abstracting, and remember that the object *is not* the event and that we have abstracted characteristics fewer than, and different from, those the event has, we should expect many unforeseen happenings to occur. Consequently, when the unexpected happens, we are saved from painful and harmful semantic shocks.

If, through lack of consciousness of abstracting, we identify or confuse words with objects and feelings, or memories and 'ideas' with experiences which belong to the un-speakable objective level, we identify higher order abstractions with lower. Since this special type of semantic identification or confusion is extremely general, it deserves a special name. I call it *objectification,* because it is generally the confusion of words or verbal issues (memories, 'ideas'.,) with objective, un-speakable levels, such as objects, or experiences, or feelings,. If we objectify, we *forget,* or we *do not remember* that words *are not* the objects or feelings themselves, that the verbal levels are always different from the objective levels. When we identify them, we disregard the inherent differences, and so proper evaluation and full adjustment become impossible.

Similar semantic difficulties arise from the confusion of higher order abstractions; for instance, the identification of inferences with

descriptions. This may be made clearer by examples. In studying these examples, it should be remembered that the organism acts as-a-whole, and that 'emotional' factors are, therefore, always present and should not be disregarded. In this study, the reader should try to put himself *'emotionally'* in the place of the Smith we speak about; then he cannot fail to understand the serious semantic disturbances these identifications create in everybody's life.

Let us begin with a Smith who knows nothing of what has been said here, and who is *not* conscious of abstracting. For him, as well as for Fido, there is, in principle, no realization of the 'characteristics left out'. He is 'emotionally' convinced that his words entirely cover the 'object' which '*is* so and so'. He identifies his lower abstractions with characteristics left out, with higher abstractions which have all characteristics included. He ascribes to words an entirely false value and certitude which they cannot have. He does not realize that his words may have different meanings for the other fellow. He ascribes to words 'emotional' *objectivity* and value, and the verbal, *A* 'permanence', 'definiteness', 'one-value'., to objects. When he hears something that he does not like, he does not ask 'what do you mean?', but, under the semantic pressure of identification, he ascribes his own meanings to the other fellow's words. For him, words *'are'* 'emotionally' overloaded, objectified semantic fetishes, even as to the primitive man who believed in the 'magic of words'. Upon hearing anything strange, his *s.r* is undelayed and may appear as, 'I disagree with you', or 'I don't believe you',. There is no reason to be dramatic about any unwelcome statement. One needs definitions and interpretations of such statements, which probably are correct from the speaker's point of view, if we grant him his informations, his *undefined terms,* the structure of his language and premises which build up his *s.r.* But our Smith, innocent of the 'structure of human knowledge', has mostly a semantic belief in the one-value, absoluteness., of things, and thinghood of words, and does not know, or does not *remember,* that words *are not* the events themselves. Words represent higher order abstractions manufactured by higher nerve centres, and objects represent lower order abstractions manufactured by lower nerve centres. Under such *identity-delusions,* he becomes an absolutist, a dogmatist, a finalist,. He seeks to establish 'ultimate truths', 'eternal verities'., and is willing to fight for them, never knowing or remembering, otherwise forgetting, the 'characteristics left out'; never recognizing that the noises he makes *are not* the objective actualities we deal with. If somebody contradicts him, he is much disturbed. Forgetting characteristics left out, he is always 'right'. For him his statement is not

only *the* only statement possible, but he actually attributes some cosmic objective evaluation to it.

The above *description* is unsatisfactory, but cannot be much improved upon, since the situation involves *un-speakable* affective components which *are not* words. We must simply try to put ourselves in his place, and to live through his experiences when he identifies and believes without question that his words *'are'* the things they only stand for. To give the full consequences of such identification resulting in wrong evaluation, I might add most tedious *descriptions* of the interplay of situations, evaluations., in quarrels, unhappinesses, disagreements., leading to dramas and tragedies, as well as to many forms of 'mental' illness effectively described only in the *belles-lettres*. Thus, $Smith_1$, who is *not* conscious of abstracting, makes the statement, 'A circle is not square'. Let us suppose that $Brown_1$ contradicts him. $Smith_1$ is angered; for his *s.r*, his statement 'is' the 'plain truth', and $Brown_1$ must be a fool. He objectifies it; ascribes to it undue value. For him, it 'is' 'experience', a 'fact'., and he bursts into speech, denouncing $Brown_1$ and showing how wrong he 'is'. From this semantic attitude, many difficulties and tragedies arise.

But if $Smith_2$ (conscious of abstracting) makes the statement, 'A circle is not square', and $Brown_2$ contradicts him, what would $Smith_2$ do? He would smile, would not burst into speech to defend *his* statement, but would ask $Brown_2$, 'What do you mean? I do not quite understand you'. After receiving some answer, $Smith_2$ would explain to $Brown_2$ that his statement is not anything to quarrel about, as it is verbal and is true only by *definition*. He would also grant the right of $Brown_2$ *not* to accept *his definition,* but to use another one to satisfy himself. The problem would then, naturally, arise as to what definition both could accept, or which would be generally acceptable. And the problem would then be solved by purely pragmatic considerations. Words appear as creatures of definitions, and optional; but this attitude involves important and new *s.r*.

This fact seems of tremendous semantic importance, as it provides the working foundation for a theory of 'universal agreement'. In the first part of the above example, $Smith_1$, according to the accepted standards, was 'right' ('a circle is not square'). Is he 'more right' than $Brown_1$, for whom the 'circle is square'? Not at all. Both statements belong to the verbal level, and represent only forms of representation for *s.r inside their skin*. Either may be 'right' by some explicit or implicit 'definitions'. Are the two statements equally valid? This *we do not know a priori;* we must investigate to find out if the noises uttered have

meanings outside of pathology, or which statement structurally covers the situation better, carries us structurally further in describing and analysing this world,. Only scientific structural analysis can give the preference to one form over another. Smith and Brown can only produce their 'definitions' according to their *s.r*, but they are *not* judges as to which 'definitions' will *ultimately* stand the test of structure.

The moment we eliminate identification we become conscious of abstracting, and permanently and instinctively remember that the object is *not* the event, that the label is *not* the object, and that a statement about a statement is *not* the first statement; thus, we reach a semantic state, where we recognize that everybody 'is right' by his own 'definitions'. But any individual or unenlightened public opinion is not the sole judge as to what 'definitions' and what language should prevail. Only structural investigation (science) can decide which appears as the structurally more similar form of representation on the verbal levels for what is going on on the un-speakable, objective levels.

When it comes to 'description of facts', the situation is not fundamentally altered. Mistakes seem always possible and often occur. Besides, the semantic impressions which 'facts' make on us are also individual, and often in conflict, as comparison of the testimonies of eye-witnesses shows. But there is no need for permanent disagreement; more structural investigation of the objective and verbal levels will provide a solution. Once such an investigation is carried far enough, we can always reach a semantic basis where all may agree, provided we do not identify, do not objectify, and do not confuse description and inference, descriptive and inferential words,.

As our analysis is carried out from the structural and *non-el* point of view, we should not miss the fact that semantic components associated with words and statements are, outside of very pathological cases, never entirely absent, and become of paramount importance. In the older days, we had no simple and effective means by which we could affect painful, misplaced, or disproportionate evaluations, meanings., through a semantic re-education, which are supplied by the present analysis and the use of the Structural Differential. The means to eliminate identification consist of: first, an *objective* relief diagram to which we can *point our finger;* and second, a convincing explanation (pointing the finger to the labels) that the verbal levels, with their distressing and disastrous older *s.r*, *are not,* and differ entirely from, the levels of objects and events. Whatever we may say or feel, the objects and events remain on the un-speakable levels and cannot be reached by words. Under such natural structural conditions, we can only reach the objective level by

seeing, handling, actually feeling . , and, therefore, by pointing our finger to the object on the Structural Differential and being silent—all of which cannot be conveyed by words *alone*.

In experiments with the 'mentally' ill in whom the semantic disturbances were very strong, it took several months to train the patients in non-identity and in silence on the objective levels. But, as soon as this was achieved, either complete or partial relief followed.

The main disturbances in daily life, as well as in 'mental' illness, are found in the affective field. We find an internal pressure of identifications, expressed by bursting into speech, and unjustified semantic over-evaluation of words, the ascribing of objectivity to words , . In such cases, suppression or repression of words does not accomplish much, but often does *considerable harm* and must be avoided by all means. Under such conditions, the use of the relief diagram becomes a necessity in pointing to the difference between different orders of abstractions and inducing the semantically beneficial silence on the 'objective level' without repression or suppression.

With the use of the Structural Differential, we can eliminate identification, and so attain the benefits, avoiding the dangers. If any one identifies, and his *s.r* drive him into an outburst of speech, we do not repress or suppress him; we say, instead: 'At your pleasure [since it makes him feel better], but remember that your words occur on the verbal levels [showing with a gesture of the hands the hanging labels], and that they *are not* the objective level, which remains untouched and unchanged'. Such a procedure, when repeated again and again, gives him the proper semantic *evaluation of orders of abstractions, frees him from identification, yet without repression or suppression.* It teaches him, also, to enquire into alleged 'facts', and then to try to find structurally better forms of representation. If such results are not forthcoming, we may use the older forms, but by proper evaluation we do not semantically put 'belief' in these forms of representation. Such beliefs always appear as the result of identification somewhere.

The technique of training is simple. We live on the 'objective' or lower order of abstraction levels, where we must see, feel, touch, *perform* . , but *never* speak. In training, we must use our hands , . It is very useful, after the Structural Differential has been repeatedly explained, stressing, in particular, the rejection of the 'is' of identity, not to interrupt the other fellow. Let him speak, but wave the hand, indicating the verbal levels; then point the finger to the objective level, and, with the other hand, close your own lips, to show that on the objective level one can only be silent. When performed repeatedly, this pantomime has a

most beneficial, semantic, pacifying effect upon the 'over-emotionalized' identification-conditions. The neurological mechanism of this action is not fully known, but some aspects are quite clear.

The more elaborate a nervous system becomes, the further some parts of the brain are removed from immediate experience. Nerve currents, having finite velocity, eventually have longer and more numerous paths to travel; different possibilities and complications arise, resulting in 'delayed action'. It is known that the thalamus (roughly) appears connected with affective and 'emotional' life, and that the cortex, farther removed and isolated from the external world, has the effect of inducing this 'delay in action'. In unbalanced and 'emotional' 'thinking', which is so prevalent, the thalamus seems overworked, the cortex seems not worked enough. The results take on the form of a low kind of animalistic, primitive, or infantile behaviour, often of a pathological character in a supposedly civilized adult. It appears that the 'silence on objective levels' introduces this 'delayed action', unloading the thalamic material on the cortex. This psychophysiological method is very simple, scientific, and entirely general. The standard 'mental' therapy of today applies also a *method of re-education* of *s.r*, as if relieving the thalamus, and putting more of the nerve currents through the cortex, or eventually furnishing the cortex with different material, so that the thalamic material returning from the cortex could be properly influenced.

If we succeed in such a semantic re-education, the difficulties vanish. The older experimental data show that in many instances we have succeeded, and that in many we have failed. The successful cases show that we actually know the essential semantic points involved; the failures show that we do not know enough, and that our older theories are not sufficiently general. At present, only the more pronounced and morbid semantic disturbances come to the attention of physicians, and very little is done by way of *preventive* measures. Besides the pronounced disturbances in daily life, we see an enormous number of semantic disturbances which we disregard, and call 'peculiarities'. In the majority of cases, these 'peculiarities' are undesirable, and, under unfavorable conditions, may lead to more serious consequences of a morbid character. They usually involve a great deal of unhappiness for all concerned, and unhappiness appears as a sign of some semantic maladjustment somewhere, and so may be destructive to 'mental' and nervous health.

In advanced 'mental' illnesses, such as usually come to the attention of psychiatrists, there are certain psycho-logical symptoms which are generally present. The symptoms of interest to us in this work are called 'delusions', 'illusions', and 'hallucinations'. All of them involve the

semantic identification or *confusions of the orders of abstractions,* the evaluation of lower orders of abstraction as higher, or higher as lower. It was explained already that some components of identification are invariably present there, and so identification may be considered as an elementary type of semantic disturbances from which all the other states differ only in intensity.

The main point is to find psychophysiological preventive means whereby this identification can be forestalled or eliminated. To date, experience and analysis show that all forms of identification may be successfully eliminated by training in *visualization,* if this semantic state can be produced. For this purpose the Structural Differential is uniquely useful and necessary. With its help we train all centres. The lower centres are involved, as we see, feel, hear.; the higher centres are equally involved, as we 'remember', 'understand'.; *with the result that all centres work together without conflict.* The 'consciousness of abstracting' is inculcated, replacing vicious *s.r* of confusion of orders of abstractions and identification.

This harmonious working of all centres on their proper levels has extremely far-reaching, practical consequences in 'mental' and physical hygiene. We become co-ordinated, adjusted, and difficulties which might otherwise occur in the future are eliminated in a preventive way. It must be remembered that, at present, it is impossible to foresee to how great an extent the elimination of identification on all levels will have a beneficial effect. At this stage we know even experimentally that the benefits are very large, but we may expect that they will become still more numerous when more experimenting has been done. Delusions, illusions, and hallucinations represent manifestations which occur in practically all 'mental' difficulties, and they only represent a semantic identification of orders of abstractions of different degrees of intensity. When this confusion is eliminated, we may expect general changes in the symptoms. But as the correspondence is probably not *one-to-one,* it is impossible to foretell theoretically what improvements may be expected in pronounced illness. In the slighter disturbances, which affect us in daily life, the results are much easier to foresee, and are *always* beneficial.

To how great an extent the consciousness of abstracting benefits semantically *the whole organism,* I may illustrate by one of my own experiences. Once I was travelling on a ship. A gentleman visited my cabin, and, seeing the Structural Differential, asked questions about it. After a short explanation, he asked about practical applications.

My guest was sitting on my berth; I was sitting on a small folding chair. I got up, went to the door, then pretended that I was coming in,

and, at my suggestion, he said, 'Please have a seat'. I remained standing while explaining how, if I were not 'conscious of abstracting', to me his word 'seat' would be identified with the chair (objectification) and my *s.r* would be such that I would sit down with great confidence. If the chair were to collapse I would have, besides the bump, an affective shock, 'fright'., which might do harm to my nervous system. But if I were conscious of abstracting, my *s.r* would be different. I would remember that the *word,* the *label* 'seat' is *not* the thing on which I am supposed to sit. I would remember that I am to sit on this individual, unique, un-speakable object, which might be strong or weak,. Accordingly, I would sit carefully. In case the chair should collapse, and I should hurt myself physically, I would still have been saved an affective nervous shock.

During all these explanations I was handling the little chair and shaking it. I did not notice that the legs were falling out, and that the chair was becoming unfit for use. Then, when I actually sat on the relic, it gave way under me. However, I did *not fall* on the floor. I caught myself in the air, so to say, and saved myself from a painful experience. It is important to notice that such physical readiness requires a very elaborate, nervous, unconscious co-ordination, which was accomplished by the semantic state of *non-identification* or *consciousness of abstracting*. When such a consciousness of abstracting is acquired, it works instinctively and automatically and does *not* require continual effort. Its operation involves a fraction of a second's *delay in action,* but this small delay is not harmful in practice; on the contrary, it has very important psycho-logical and neurological 'delayed action' effects.

It seems that 'silence' on the objective levels involves this psycho-physiological delay. No matter how small, it serves to unload the thalamic material on the cortex. In a number of clinical cases, Dr. Philip S. Graven has demonstrated that the moment such a delay can actually be produced in the patient, he either improves or is entirely relieved. The precise neurological mechanism of this process is not known, but there is no doubt that this 'delayed action' has many very beneficial effects upon the whole working of the nervous system. It somehow balances harmful *s.r*, and also somehow stimulates the higher nervous centres to more *physiological* control over the lower centres.

A very vital point in this connection should be noticed. That this 'delayed action' is beneficial is acknowledged by the majority of normally developed adults in the form of delay in action and finds its expression in such statements as 'think twice', 'keep your head', 'hold your horses', 'keep cool', 'steady', 'wait a minute'., and such functional

recipes as 'when angry, count ten',. In daily life, such wisdom is acquired either by painful experience, or is taught to children in an *A* language, which, as practice shows, is rarely effective because of its inadequacy. It is seldom realized that the mechanism of these functional observations and familiar advices have very powerful and workable underlying *neurological processes,* which can be *reached and directly affected by psychophysiological,* ordinal, *non-el* methods in connection with *the structure of the language we use.* Thus, under an infantile, *A*, and prevailing system we use and teach our children a language involving the 'is' of identity, and so we must confuse orders of abstractions, preparing for ourselves and the children the harmful semantic predispositions for 'bursting into speech', instead of 'wait a minute', which, neurologically, means abusing our thalamus and keeping our cortex 'unemployed'. In a $\bar{A}$ ∞-valued system we reject the 'is' of identity; we cannot confuse orders of abstractions; we cannot identify words with the un-speakable objective levels or inferences with descriptions, and we cannot identify the different abstractions of different individuals,. This semantic state of proper evaluation results in discrimination between the different orders of abstractions; an automatic delay is introduced—the cortex is switched completely into the nervous circuit. The semantic foundation is laid for 'higher mentality' and 'emotional balance'.

We have already had occasion to mention the mechanism of projection in connection with identification, as a semantic state of affective ascribing of lower centre characteristics to higher order abstractions and vice versa, and in connection with the introverted or extroverted attitudes. Likewise, we have already reached the conclusion that a well-adjusted and, therefore, well-balanced individual should be neither of the extremes, but a balanced extroverted introvert. By training with the Differential this important semantic result may be brought about. By training with the 'object' on its level, we become extroverted, and we learn to observe; this results in semantic freedom from 'preconceived ideas', such as we have when we start with the evaluation, label first and object next, instead of the natural order, object first and label next. By passing to higher order abstractions and evaluating the successive ranks of labels, we train in introversion. The result, as a whole, is that we may achieve the desirable and balanced semantic state of the extroverted introvert.

That in the training with the Differential we use all available nerve centres is beneficial, because the lower centres are in closer connection with the vegetative nervous system than are the others.

CHAPTER XXVII

HIGHER ORDER ABSTRACTIONS

The characters which science discerns in nature are subtle characters, not obvious at first sight. They are relations of relations and characters of characters. (573)

A. N. WHITEHEAD

In this connection one should particularly remember that the human language permits the construction of sentences which do not involve any consequences and which therefore have no content at all—in spite of the fact that these sentences produce some kind of picture in our imagination; e. g., the statement that besides our world there exists another world, with which any connection is impossible in principle, does not lead to any experimental consequence, but does produce a kind of picture in the mind. Obviously such a statement can neither be proved nor disproved. One should be especially careful in using the words "reality," "actually," etc., since these words very often lead to statements of the type just mentioned. (215)

W. HEISENBERG

Section A. General.

In the previous chapters I demonstrated that there is a short cut which enables us to grasp, acquire, and apply what has been advanced in the present work. This semantic short cut is 'consciousness of abstracting'. It is a psycho-logical attitude toward all our abstracting on all levels, and so involves the co-ordinated working of the organism-as-a-whole.

The use of the Structural Differential is necessary, because some levels are un-speakable. We can see them, handle them, feel them., but under *no* circumstances can we reach those levels by speech alone. We must, therefore, have a diagram, by preference in relief form, which represents the empirical structural conditions, and which indicates the un-speakable level by some other means than speech. We must, in the simplest case, either point our finger to the object, insisting upon silence, or must perform bodily some activity and similarly insist upon silence, as the performing and feelings are also *not* words.

In such semantic training it is enough to insist upon the non-identity or the difference between the objective, *un-speakable* levels of lower order abstractions, (O_h), and the verbal or higher order abstractions, (L_n). When this habit and feeling are acquired, no one should have difficulties in extending the non-identity method to daily-life occurrences. To achieve these semantic aims, we must first emphasize the common-sense fact that an object *is not* the event. To do this, we start with the 1933 scientific structural 'metaphysics' about the event and

stress the fact that the object, being a nervous abstraction of lower order, has fewer and different *m.o* characteristics than the event has. This is best accomplished by stressing the fact that in abstracting from the event to the object we left out some characteristics. We did not *abstract 'all'* characteristics; this would be a self-contradiction in terms, an impossibility.

We do not even need to stress a full understanding of the event. Common-sense examples, showing that what we recognize as a 'pencil' is not 'all', often suffice. No one will have difficulties, provided he trains himself in this direction, in *remembering* continually and instinctively the free hanging strings (B′), (B″), which indicate the non-abstracted or left-out characteristics and which help to train in *non-identity*. With the relief diagram, the *s.r* of the student are trained *through all nervous centres*. He sees, he handles., the hanging strings, and he also hears about them. This gives the *maximum probability* that the organism-as-a-whole will be affected. In this way an 'intellectual' theory engages the 'senses', feelings, and reflex mechanisms. To affect the organism-as-a-whole, organism-as-a-whole methods must be employed.

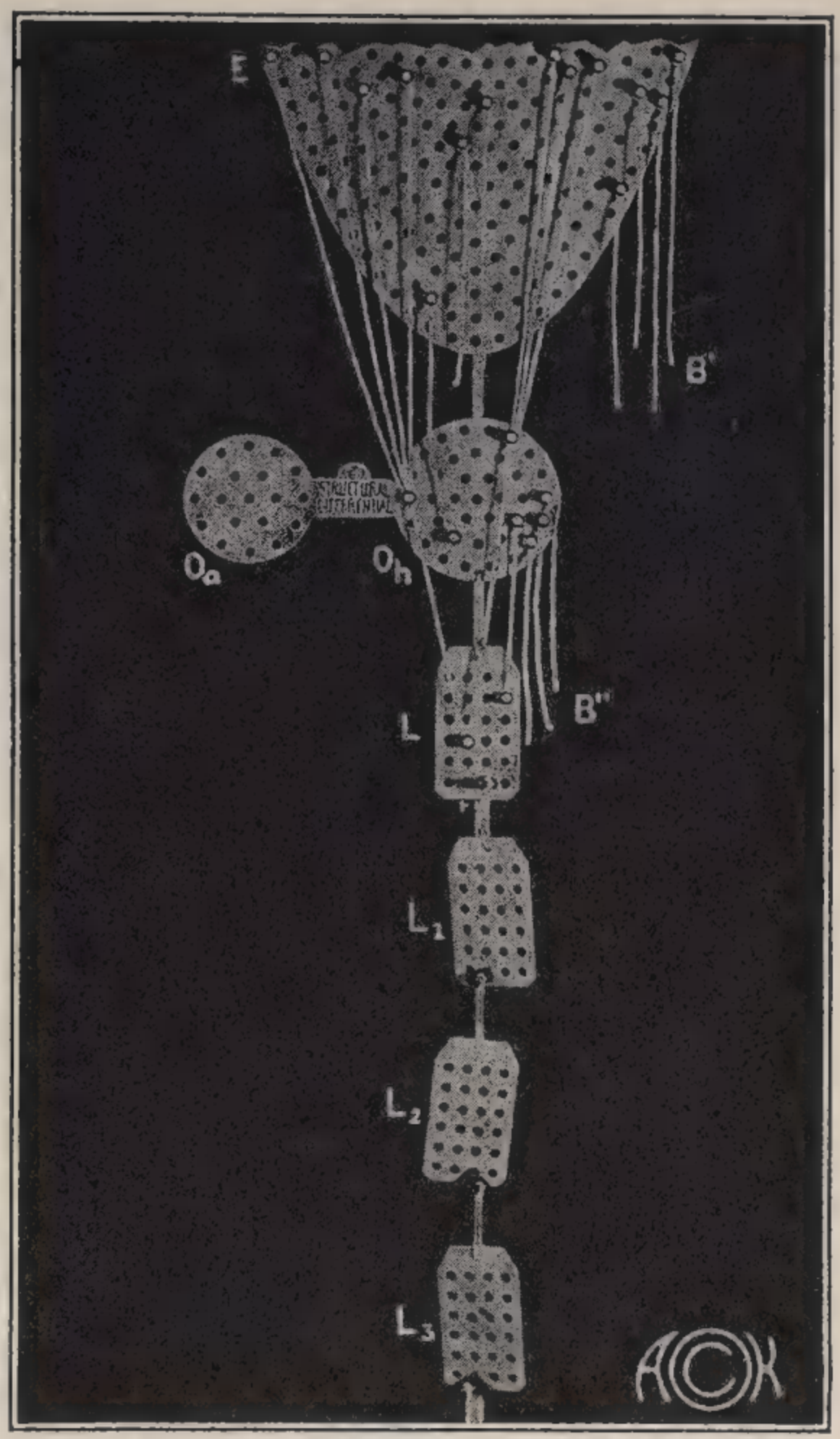


FIG. 1
THE STRUCTURAL DIFFERENTIAL

A similar structural situation is found when we deal with higher order abstractions. A word, or a name, or a statement is conveyed in spoken form or by writing, and affects first the lower centres and then is abstracted, and again transformed, by the higher centres. The order

is generally not changed when the verbal issues are neither seen nor heard but originate in ourselves. Most 'impulses', 'interests', 'meanings', 'evaluations'., originate in lower centres and follow the usual course, from lower centres to higher. When 'experience' (reaction of lower centres) is transformed into 'memories' (higher centres)., the order is similar. Difficulties begin when the order is pathologically *reversed* and 'ideas' are evaluated as experience, words as objects,. In the building of language a similar process can be observed. We observe the absolute individuals with which we actually deal, we label them with individual names, say, $A_1, A_2, \ldots, A_{11}, A_{12}, \ldots, A_{21}, A_{22}, \ldots, A_{31}, A_{32}, \ldots,$. By a process of abstracting and *disregarding,* for instance, the characteristic subscripts '1', we would have only the ones which have the characteristic subscripts 2, 3, . . . , 9, 22, 23, . . . , 29,. *Disregarding* the characteristic subscripts '2', we would have the ones with the characteristics 3, 4, . . . , 9, 33, 34, . . . , 39,. Finally, if we should eliminate all *individual* characteristic subscripts, we would have a 'general' name A for the whole group without singling out individual characteristics.

All words of the type of 'man', 'animal', 'house', 'chair', 'pencil'., have been built by a similar process of abstraction, or *disregard* for individual differences. In each case of disregard of individual characteristics a *new* neurological process was involved.

Similarly, with 'statements about a statement'. When we hear a statement, or see it in a written form, such a statement becomes a stimulus entering through the lower centres, and a statement about it represents, in general, a new process of abstraction, or an abstraction of higher order.

It becomes obvious that the introduction of a language of 'different order of abstractions', although it is not familiar, yet structurally it represents very closely, in terms of *order,* most fundamental neurological processes going on in us. As we already know, a natural order has been established by evolution; namely, lower order abstractions first, higher next; the identifications of orders or the reversal of orders appears pathological for man and appears as a confusion of orders of abstractions, resulting in *false evaluation:* identification, illusions, delusions, and hallucinations.

Historically, the first to pay serious attention to the above problems in a consistent, yet very limited, way were mathematicians. In the investigation of the problems of the foundation of mathematics, mathematical 'logic', and the theory of aggregates, we came across self-contradictions which would make mathematics impossible. To avoid such a disaster, Russell invented what is called the 'theory of mathematical

types'. The status of this theory is a very interesting and instructive one. The theory solves the mathematical difficulties, thus saving mathematics, but has *no* application to life. Practically all mathematicians, if I am not mistaken, the author of the theory included, somehow 'dislike' the theory and make efforts to solve the problems in a different way and possibly to abandon the theory altogether.

We have already shown that the introduction of a language of 'different orders of abstractions' is structurally entirely justified and physiologically natural, as it describes, in terms of order, the activities of the nervous system. Such facts are important; but if, in addition, the introduction of a language of a new $\bar{A}$ structure would give us further demonstrable advantages, then the introduction of such a language would become increasingly desirable.

Although the majority of mathematicians 'dislike' the theory of types, yet, at present, this theory is unconditionally necessary for non-self-contradictory mathematics. The author was pleasantly surprised to find that after his $\bar{A}$-system was formulated, this simple and natural, actional, functional, operational, *non-el* theory covers the theory of mathematical types and generalizes it, making the theory applicable not only to the solution of mathematical paradoxes but to the solution of the majority of purely human and scientific difficulties. One general rule of 'non-confusion of orders of abstractions', and the acquiring of the simple and workable 'consciousness of abstracting' based on the denial of the 'is' of identity, offers a *full* structural and semantic solution. The disregard of the issues involved leads fatalistically to the manufacture of endless and unnecessary human sufferings and unhappiness, the elimination of which is one of the main points in a theory of sanity. There is no mystery in 1933 that continuous small painful shocks may lead to serious semantic and physical disturbances. Psycho-logicians and psychiatrists will find it increasingly difficult to work at their problems if they disregard these semantic issues. Parents and teachers will find simple yet effective structural means for training the reactions of children *in sanity,* with all the ensuing semantic benefits to the individuals and to society.

When Whitehead and Russell were working at the foundations of mathematics, they came across endless paradoxes and self-contradictions, which, of course, would make mathematics impossible. After many efforts they found that all these paradoxes had one general source, in the rough, in the expressions which involve the word 'all', and the solution was found by introducing 'non-allness', a semantic forerunner of non-identity. Consider, for example, 'a proposition about *all* proposi-

tions'. They found that such totalities, or such 'all' statements, were not legitimate, as they involved a self-contradiction to start with. A proposition cannot be made legitimately about 'all' propositions without some restriction, since it would have to include the new proposition which is being made. If we consider a *m.o* term like 'propositions', which we can manufacture without known limits, and remember that any statement *about* propositions takes the form of a proposition, then obviously we cannot make statements about *all* propositions. In such a case the statement must be limited; such a set has *no total,* and a statement about 'all its members' cannot be made legitimately. Similarly, we cannot speak about *all* numbers.

Statements such as 'a proposition about *all* propositions' have been called by Russell 'illegitimate totalities'. In such cases, it is necessary to break up the set into smaller sets, each of which is capable of having a totality. This represents, in the main, what the theory of types aims to accomplish. In the language of the *Principia Mathematica,* the principle which enables us to avoid the illegitimate totalities may be expressed as follows: 'Whatever involves *all* of a collection must not be one of the collection', or, 'If, provided a certain collection had a total, it would have members only definable in terms of that total, then the said collection has no total'.[1] The above principle is called the 'vicious-circle principle', because it allows us to evade the vicious circles which the introduction of illegitimate totalities involve. Russell calls the arguments which involve the vicious-circle principle, 'vicious-circle fallacies'.

As an example, Russell gives the two-valued law of 'excluded third', formulated in the form that 'all propositions are true or false'. We involve a vicious-circle fallacy if we argue that the law of excluded third takes the form of a proposition, and, therefore, may be evaluated as true or false. Before we can make any statement about 'all propositions' legitimate, we must limit it in some way so that a statement about this totality must fall outside this totality.

Another example of a vicious-circle fallacy may be given as that of the imaginary sceptic who asserts that he knows nothing, but is refuted by the question—does he *know* that he *knows* nothing? Before the statement of the sceptic becomes significant, he must limit, somehow, the number of facts concerning which he asserts his ignorance, which represent an illegitimate totality. When such a limitation is imposed, and he asserts that he is ignorant of an extensional series of propositions, of which the proposition about his ignorance is not a member, then such scepticism cannot be refuted in the above way.

We do not need to enter into further details concerning the elaborate and difficult theory of types. In my $\bar{A}$ psychophysiological formulation, the theory becomes structurally extremely simple and natural, and applies to mathematics as well as to a very large number of daily experiences, eliminating an unbelievably large number of misunderstandings, vicious circles, and other semantic sources of human disagreements and unhappiness.

It should be noticed that, in the given examples, we always made a statement *about* another statement, and that the vicious circle arose from identifying or from the confusion of the orders of statements. The way out is found in the consciousness of abstracting, which leads to the semantic discrimination between *orders of abstractions.* If we have certain propositions, $p_1, p_2, p_3, \ldots p_n$, and make a new proposition about these propositions, say P, then, according to the present theory, the statement P about the statements $p_1, p_2.$, must be considered as an abstraction of higher order, and so different, and must not be identified as to order with the propositions $p_1, p_2, \ldots p_n$.

The above psychophysiological formulation is entirely general, yet simple and natural in a $\bar{A}$-system. To make this clearer, I shall take several statements concerning the theory of types from the *Principia Mathematica,* shall designate them by (Pr.), shall reformulate them in my language of *orders of abstractions,* and shall designate them as general semantics (G. S.).

Thus, 'The vicious circles in question arise from supposing that a collection of objects may contain members which can only be defined by means of the collection as a whole' (Pr.). Objects as individuals and 'collections of objects' obviously belong to different orders of abstractions and should not be confused (G.S.). A 'Proposition about *all* propositions' (Pr.). This involves a confusion of orders of abstractions, for if we posit propositions $p_1, p_2, \ldots p_n$, then a proposition P *about* these propositions represents a higher order abstraction and should not be identified with them (G.S.). 'More generally, given any set of objects such that, if we suppose the set to have a total, it will contain members which presuppose this total, then such a set cannot have a total. By saying that a set has "no total", we mean, primarily, that no significant statement can be made about "all its members"' (Pr.). A set of statements or objects or elements, or the like, and a statement *about* them belong to different orders of abstractions and should not be confused (G.S.). In the language of Wittgenstein: 'No proposition can say anything about itself, because the propositional sign cannot be contained in itself (that is the "whole theory of types").'[2]

In the language of the present general semantics a statement about a statement is not the 'same' statement, but represents, by structural and neurological necessity, a higher order of abstraction, and should not be confused with the original statement.

Similar reformulations apply to all cases given in the *Principia Mathematica,* and so it becomes evident that the present theory covers a similar ground as the theory of types, and also covers an endless list of daily-life applications which are of crucial semantic importance in a theory of sanity. We must stress here a simple, natural, and *single semantic law of non-identity which covers all confusions of orders of abstraction.* This one rule and training teach us not to confuse the higher orders with the lower, not to identify words with objects (not to objectify), as well as not to confuse higher abstractions of different orders. This generality and structural simplicity constitute an argument in favor of the present $\bar{A}$-system. It is easier to teach a single, simple, and natural rule which covers a vast field of semantic sources of human difficulties. For when the rule is explained, and the learner is trained with the Structural Differential, the semantic problem resolves itself simply into the showing with one's finger different orders of abstractions, and insisting that 'this *is not* this'.

If we consider the natural, structural, and *empirical* fact that our lives are lived in a world of non-identical abstractions of different orders, the discrimination between different orders becomes of paramount semantic importance for evaluation. Under such conditions we should become thoroughly acquainted with the mechanism of these different orders of abstractions. We should notice, first, that the language of the *Principia Mathematica* is A, and involves the 'is' of identity,. Such a language leads to identifications and to confusions, and makes simple issues difficult and perplexing. The term 'class' is very confusing. What do we mean by this term? In life we have, and deal with, *individuals* on *objective, un-speakable* levels. If we take a number of individuals, we have a number of them, yet they all remain individual. If we produce an abstraction of higher order, so that the individuality of each member is lost, then we have an abstraction of a higher order ('idea' in the old language), but no more the absolute individuals of our collection. The term 'class' in this respect is seriously confusing, as it tends to conceal a simple experimental fact, and leads to confusion of the orders of abstractions if the *multiordinality* of the term 'class' is not formulated.

Many critics and reviewers of the *Principia Mathematica* somehow feel this to be so, but their criticisms are not bold enough, and do not

go to the roots of the *A* semantic difficulty. They do not pay attention to the *A*, 'logical', 'philosophical', and 'psychological' *elementalistic* method and language involving the 'is' of identity, in which the Introduction of the *Principia* is written. Doctor Alonzo Church is the first, as far as my knowledge goes, to suggest that, following Peano, numbers should be defined in the language of abstractions. He does not carry his analysis further, however, and does not state that it involves a language of entirely different $\bar{A}$ structure.[3] If we abandon the term 'class' and accept the language of 'abstractions of *different* orders', then we are led to the rejection of the 'is' of identity and to the present system, of which the theory of mathematical types becomes a necessary part. The problems of 'class' cease to be an 'assumption', as the different orders of abstractions are descriptions of experimental facts; and so the 'axiom of reducibility' becomes unnecessary. In my language, this axiom is also an *aristotelian description* of the experimental fact that we can abstract in different orders.

Section B. Multiordinal terms.

In the examples given in Section A, we used words such as 'proposition', which were applied to all higher order abstractions. We have already seen that such terms may have different uses or meanings if applied to different orders of abstractions. Thus originates what I call the *multiordinality* of terms. The words 'yes', 'no', 'true', 'false', 'function', 'property', 'relation', 'number', 'difference', 'name', 'definition', 'abstraction', 'proposition', 'fact', 'reality', 'structure', 'characteristic', 'problem', 'to know', 'to think', 'to speak', 'to hate', 'to love', 'to doubt', 'cause', 'effect', 'meaning', 'evaluation', and an endless array of the most important terms we have, must be considered as *multiordinal terms.* There is a most important semantic characteristic of these *m.o* terms; namely, that they are ambiguous, or ∞-valued, in general, and that each has a definite meaning, or one value, only and exclusively in a given context, when the order of abstraction can be definitely indicated.

These issues appear extremely simple and general, a part and parcel of the structure of 'human knowledge' and of our language. We cannot avoid these semantic issues, and, therefore, the only way left is to face them explicitly. The test for the multiordinality of a term is simple. Let us make any statement and see if a given term applies to it ('true', 'false', 'yes', 'no', 'fact', 'reality', 'to think', 'to love',.). If it does, let us deliberately make another statement *about* the former statement and test if the given term may be used again. If so, it is a safe assertion that this term should be considered as *m.o.* Any one can test such a *m.o*

term by himself without any difficulty. The main point about all such *m.o* terms is that, *in general,* they are *ambiguous,* and that all arguments about them, 'in general', lead only to *identification of orders of abstractions and semantic disturbances, and nowhere else.* Multiordinal terms have only definite meanings on a given level and in a given context. Before we can argue about them, we must fix their orders, whereupon the issues become simple and lead to agreement. As to 'orders of abstraction', we have no possibility of ascertaining the 'absolute' order of an abstraction; besides, we *never* need it. In human semantic difficulties, in science, as well as in private life, usually no more than three, perhaps even two, neighbouring levels require consideration. When it comes to a serious discussion of some problem, errors, ambiguity, confusion, and disagreement follow from confusing or identifying the neighbouring levels. In practice, it becomes *extremely simple* to settle these three (or two) levels and to keep them separated, *provided we are conscious of abstracting, but not otherwise.*

For a theory of sanity, these issues seem important and structurally essential. In identifications, delusions, illusions, and hallucinations, we have found a *confusion* between the orders of abstractions or a false evaluation expressed as a reversal of the natural order.

One of the symptoms of this confusion manifests itself as 'false beliefs', which again imply comparison of statements about 'facts' and 'reality', and involve such terms as 'yes', 'no', 'true', 'false',. As all these terms are multiordinal, and, therefore, ambiguous, 'general' 'philosophical' rigmaroles should be avoided. With the consciousness of abstracting, and, therefore, with a *feel* for this peculiar stratification of 'human knowledge', all semantic problems involved can be settled simply.

The avoidance of *m.o* terms is impossible and undesirable. Systematic ambiguity of the most important terms follows systematic analogy. They appear as a direct result and condition of our powers of abstracting in different orders, and allow us to apply one chain of ∞-valued reasoning to an endless array of different one-valued facts, all of which are different and become manageable only through our abstracting powers.

For further details about the theory of types, the reader is referred to the literature on the subject and Supplement II[4]; here I shall give only a few examples of the complexities and difficulties inherent in language, and show how simply they become solved by the aid of $\bar{A}$ general semantics and the resulting 'consciousness of abstracting'.

As an example, I quote Russell's analysis of the 'simple' statement 'I am lying', as given in the *Principia.* 'The oldest contradiction of the

kind in question is the *Epimenides*. Epimenides the Cretan said that all Cretans were liars, and all other statements made by Cretans were certainly lies. Was this a lie? The simplest form of this contradiction is afforded by the man who says "I am lying"; if he is lying, he is speaking the truth, and vice versa. . . .

'When a man says "I am lying", we may interpret his statement as: "There is a proposition which I am affirming and which is false." That is to say, he is asserting the truth of some value of the function "I assert p, and p is false." But we saw that the word "false" is ambiguous, and that, in order to make it unambiguous, we must specify the order of falsehood, or, what comes to the same thing, the order of the proposition to which falsehood is ascribed. We saw also that, if p is a proposition of the nth order, a proposition in which p occurs as an apparent variable is not of the nth order, but of a higher order. Hence the kind of truth or falsehood which can belong to the statement "there is a proposition p which I am affirming and which has falsehood of the nth order" is truth or falsehood of a higher order than the nth. Hence the statement of Epimenides does not fall within its own scope, and therefore no contradiction emerges.

'If we regard the statement "I am lying" as a compact way of simultaneously making all the following statements: "I am asserting a false proposition of the first order," "I am asserting a false proposition of the second order," and so on, we find the following curious state of things: As no proposition of the first order is being asserted, the statement "I am asserting a false proposition of the first order" is false. This statement is of the second order, hence the statement "I am making a false statement of the second order" is true. This is a statement of the third order, and is the only statement of the third order which is being made. Hence the statement "I am making a false statement of the third order" is false. Thus we see that the statement "I am making a false statement of order $2n+1$" is false, while the statement "I am making a false statement of order $2n$" is true. But in this state of things there is no contradiction.'[5]

Clearly, if we should apply the language of orders of abstractions to the above case, a similar outcome is reached more generally and more simply. If we should confuse the orders of abstractions, we might naturally have an endless argument at hand. This example shows how a confusion of orders of abstractions might lead to insoluble verbal problems, and how semantically important it is that we should not identify, and that we should be conscious of abstracting, with the resulting instinctive feeling for this peculiar structural stratification of 'human

knowledge'. We should notice that with the confusion of orders of abstractions, and by the use of *m.o* terms, *without realizing their ∞-valued* character, we may always construct an endless array of such verbal arguments to befog the issues, but that as soon as we assign a definite order to the *m.o* terms, and so settle a specific single meaning in a given context for the many meanings any *m.o* term may have, the difficulties vanish.

As the above analysis applies to all *m.o* terms, and these terms happen to be most important in our lives, there is no use in trying to avoid these terms and the consequences of using them. Quite the contrary; often it is structurally necessary to build a *m.o* term—for instance, 'abstracting'—we must take for granted that it has many meanings, and indicate these meanings by assigning to the term the definite order of abstraction. Thus, such a term as 'abstracting' or 'characteristic'., might be confusing and troublesome; but 'abstracting in different orders'., is not, as in a given context we may always assign the definite order and single meaning to the term.

It has been repeatedly said that a *m.o* term has, by structural necessity, many meanings. No matter how we define it, its definition is again based on other *m.o* terms. If we try to give a *general* 'meaning' to a *m.o* term, which it cannot have, further and deeper analysis would disclose the multiordinality of the terms by which it is defined, restoring once more its multiordinality. As there is no possibility of avoiding the above structural issue, it is more correct and also more expedient to recognize at once the fundamental multiordinality of a term. If we do so, we shall not get confused as to the meaning of such a term in a given context, because, in principle, in a context its meaning is single and fixed by that context.

The semantic benefits of such a recognition of multiordinality are, in the main, sevenfold: (1) we gain an enormous economy of 'time' and effort, as we stop 'the hunting of the snark', usually called 'philosophy', or for a one-valued general definition of a *m.o* term, which would not be formulated in other *m.o* terms; (2) we acquire great versatility in expression, as our most important vocabulary consists of *m.o* terms, which can be extended indefinitely by assigning many different orders and, therefore, meanings; (3) we recognize that a definition of a *m.o* term must, by necessity, represent not a proposition but a propositional function involving variables; (4) we do not need to bother much about formal definitions of a *m.o* term outside of mathematics, but may use the term freely, realizing that its unique, in principle, meaning in a given context is structurally indicated by the context; (5) under such struc-

tural conditions, the freedom of the writer or speaker becomes very much accentuated; his vocabulary consists potentially of infinite numbers of words, and psycho-logical, semantic blockages are eliminated; (6) he knows that a reader who understands that ∞-valued mechanism will never be confused as to the meaning intended; and (7) the whole linguistic process becomes extremely flexible, yet it preserves its essential extensional one-valued character, in a given case.

In a certain sense, such a use of *m.o* terms is to be found in poetry, and it is well known that many scientists, particularly the creative ones, like poetry. Moreover, poetry often conveys in a few sentences more of lasting values than a whole volume of scientific analysis. The free use of *m.o* terms without the bother of a structurally impossible formalism outside of mathematics accomplishes this, *provided we are conscious of abstracting; otherwise only confusion results.*

It should be understood that I have no intention of condemning formalism. Formalism of the most rigorous character is an extremely important and valuable discipline (mathematics at present); but formalism, as such, in experimental science and life appears often as a handicap and not as a benefit, because, in empirical science and life, we are engaged in exploring and discovering the unknown structure of the world as a means for structural adjustment. The formal elaboration of some language is only the consistent elaboration of its structure, which must be accomplished independently if we are to have means to compare verbal with empirical structures. From a $\bar{A}$ point of view, both issues are equally important in the search for structure.

Under such structural empirical conditions the *m.o* terms acquire great semantic importance, and perhaps, without them, language, mathematics, and science would be impossible. As soon as we understand this, we are forced to realize the profound structural and semantic difference between the A and $\bar{A}$ systems. What in the old days were considered propositions, become propositional functions, and most of our doctrines become the doctrinal functions of Keyser, or system-functions, allowing multiple interpretations.

Terms belong to verbal levels and their meanings *must* be given by definitions, these definitions depending on undefined terms, which consist always, as far as my knowledge goes, of *m.o* terms. Perhaps it is necessary for them to have this character, to be useful at all. When these structural empirical conditions are taken into account, we must conclude that the postulational method which gives the structure of a given doctrine lies at the foundation of all human linguistic performances, in daily life as well as in mathematics and science. The study of these prob-

lems throws a most important light on all mysteries of language, and on the proper use of this most important human neurological and semantic function, without which sanity is impossible.

From a structural point of view, postulates or definitions or assumptions must be considered as those relational or multi-dimensional order structural assumptions which establish, conjointly with the undefined terms, the structure of a given language. Obviously, to find the structure of a language we must work out the given language to a system of postulates and find the minimum of its (never unique) undefined terms. This done, we should have the structure of such a system fully disclosed; and, with the structure of the language thoroughly known, we should have a most valuable tool for investigating empirical structure by predicting verbally, and then verifying empirically.

To pacify the non-specialist, let me say at once that this work is very tedious and difficult, although a crying need; nevertheless, it may be accomplished by a single individual. Because of the character of the problem, however, when this work is done, the semantic results have always proved thus far—and probably will continue so—quite simple and comprehensible to the common sense, even of a child.

One very important point should be noted. Since language was first used by the human race, the structural and related semantic conditions disclosed by the present analysis *have not been changed,* as they are inherent in the structure of 'human knowledge' and language. Historically, we were always most interested in the immediacy of our daily lives. We began with grunts symbolizing this immediacy, and we never realize, even now, that these historically first grunts were the most complex and difficult of them all. Besides these grunts, we have also developed others, which we call mathematics, dealing with, and elaborating, a language of numbers, or (as I define it semantically) a language of *two symmetrical and infinitely many asymmetrical unique, specific relations* for exploring the structure of the world, which is, at present, the most effective and the simplest language yet formed. Only in 1933, after many hundreds of thousands of years, have the last mentioned grunts become sufficiently elaborate to give us a sidelight on structure. We must revise the whole linguistic procedure and structure, and gain the means by which to disclose the structure of 'human knowledge'. Such semantic means will provide for the proper handling of our neurological structure, which, in turn, is the foundation for the structurally proper use of the human nervous system, and will lead to human nervous adjustment, appropriate *s.r*, and, therefore, to sanity.

Human beings are quite accustomed to the fact that words have different meanings, and by making use of this fact have produced some rather detrimental speculations, but, to the best of my knowledge, the structural discovery of the multiordinality of terms and of the psycho-physiological importance of the treatment of orders of abstractions resulting from the rejection of the 'is' of identity—as formulated in the present system—is novel. In this mechanism of multiordinality, we shall find an unusually important structural problem of human psychologics, responsible for a great many fundamental, desirable, undesirable, and even morbid, human characteristics. The full mastery of this mechanism is only possible when it is formulated, and leads automatically to a possibility of a complete psychophysiological adjustment. This adjustment often reverses the psycho-logical process prevailing at a given date; and this is the foundation, among others, of what we call 'culture' and 'sublimation' in psychiatry.

Let me recall that one of the most fundamental functional differences between animal and man consists in the fact that no matter in how many orders the animal may abstract, its abstractions stop on some level beyond which the animal cannot proceed. Not so with man. Structurally and potentially, man can abstract in indefinitely many orders, and no one can say legitimately that he has reached the 'final' order of abstractions beyond which no one can go. In the older days, when this semantic mechanism was not made structurally obvious, the majority of us copied animals, and stopped abstracting on some level, as if this were the 'final' level. In our semantic training in language and the 'is' of identity given to us by our parents or teachers or in school, the multiordinality of terms was never suspected, and, although the human physiological mechanism was operating all the while, we used it on the conscious level in the animalistic way, which means ceasing to abstract at some level. Instead of being told of the mechanism, and of being trained consciously in the fluid and dynamic *s.r* of *passing to higher and higher abstractions as normal,* for Smith, we preserved a sub-normal, animalistic semantic blockage, and 'emotionally' stopped abstracting on some level.

Thus, for instance, if, as a result of life, we come to a psychological state of hate or doubt, and stop at that level, then, as we know from experience, the lives of the given individual and of those close to him are not so happy. But a hate or doubt of a higher order reverses or annuls the first order semantic effect. Thus, hate of hate, or doubt of doubt—a second order effect—has reversed or annulled the first order effect, which was detrimental to all concerned because it remained a *structurally-stopped or an animalistic* first order effect.

The whole subject of our human capacity for higher abstracting without discernible limits appears extremely broad, novel, and unanalysed. It will take many years and volumes to work it out; so, of necessity, the examples given below will be only suggestive and will serve to illustrate roughly the enormous power of the $\bar{A}$ methods and structure, aiming to make them workable as an educational, powerful, semantic device.

Let us take some terms which may be considered as of a positive character and represent the structure of 'culture', science, and what is known in psychiatry as 'sublimation'; such as curiosity, attention, analysis, reasoning, choice, consideration, knowing, evaluation,. The first order effects are well known, and we do not need to analyse them. But if we transform them into second order effects, we then have curiosity of curiosity, attention of attention, analysis of analysis, reasoning about reasoning (which represents science, psycho-logics, epistemology.,); choice of choice (which represents freedom, lack of psycho-logical blockages, and shows, also, the semantic mechanism of eliminating those blocks); consideration of consideration gives an important cultural achievement; knowing of knowing involves abstracting and structure, becomes 'consciousness', at least in its limited aspect, taken as consciousness of abstracting; evaluation of evaluation becomes a theory of sanity,.

Another group represents morbid semantic reactions. Thus the first order worry, nervousness, fear, pity., may be quite legitimate and comparatively harmless. But when these are of a higher order and identified with the first order as in worry about worry, fear of fear., they become morbid. Pity of pity is dangerously near to self-pity. Second order effects, such as belief in belief, makes fanaticism. To know that we know, to have conviction of conviction, ignorance of ignorance., shows the mechanism of dogmatism; while such effects as free will of free will, or cause of cause., often become delusions and illusions.

A third group is represented by such first order effects as inhibition, hate, doubt, contempt, disgust, anger, and similar semantic states; the *second order reverses and annuls* the first order effects. Thus an inhibition of an inhibition becomes a positive excitation or release (see Part VI); hate of hate is close to 'love'; doubt of doubt becomes scientific criticism and imparts the scientific tendency; the others obviously reverse or annul the first order undesirable *s.r.*

In this connection the pernicious effect of identification becomes quite obvious. In the first and third cases beneficial effects were *prevented,* because identification of orders of abstractions, as a semantic

state, produced a semantic blockage which did not allow us to pass to higher order abstractions; in the second case, it actually produced morbid manifestations.

The consciousness of abstracting, which involves, among others, the full instinctive semantic realization of non-identity and the stratification of human knowledge, and so the multiordinality of the most important terms we use, solves these weighty and complex problems because it gives us structural methods for semantic evaluation, for orientation, and for handling them. By passing to higher orders these states which involve inhibition or negative excitation become reversed. Some of them on higher levels become culturally important; and some of them become morbid. Now consciousness of abstracting in all cases gives us the semantic *freedom* of all levels and so helps *evaluation* and selection, thus removing the possibility of remaining animalistically fixed or blocked on any one level. Here we find the mechanism of the 'change of human nature' and an assistance for persons in morbid states to revise by themselves their own afflictions by the simple realization that the symptoms are due to identifying levels which are essentially different, an unconscious jumping of a level or of otherwise confusing the orders of abstractions. Even at present all psychotherapy is unconsciously using this mechanism, although, as far as I know, it has never before been structurally formulated in a general way.

It should be added that the moment we eliminate identification and acquire the consciousness of abstracting, as explained in the present system, we have already acquired the permanent semantic feeling of this peculiar *structural stratification* of human knowledge which is found in the psycho-logics of the differential and integral calculus and mathematics, similar in structure to the world around us, without any difficult mathematical technique. Psycho-logically, both mathematics and the present system appear structurally similar, not only to themselves, but also to the world and our nervous system; and at this point it departs very widely from the older systems.

Let me give another example of how the recognition of order of abstractions clears up semantic difficulties.

I recall vividly an argument I had with a young and very gifted mathematician. Our conversation was about the geometries of Euclid and Lobatchevski, and we were discussing the *dropping* and *introduction* of assumptions. I maintained that Lobatchevski *introduced* an assumption; he maintained that Lobatchevski *dropped* an assumption. On the surface, it might have appeared that this is a problem of 'fact' and not of *preference*. The famous fifth postulate of Euclid reads, 'If a straight

line falling on two straight lines makes the interior angles on the same side less than two right angles, the two straight lines, if produced indefinitely, meet on that side on which are the angles less than two right angles'. We should note, in passing, that a straight line is *assumed* to be of 'infinite' length, which involves a definite type of structural metaphysics of 'space', common to the *A* and older systems. This postulate of Euclid can be expressed in one of its equivalent forms, as, for instance, 'Through a point outside a straight line one, and only one, parallel to it can be drawn'. Lobatchevski and others decided to build up a geometry *without* this postulate, and in this they were successful. Let us consider what Lobatchevski did. For this, we go to a deeper level—otherwise, to a higher order abstraction—where we discover that what on *his level* had been the *dropping* of an assumption becomes on our deeper level or higher order abstraction the *introduction* of an assumption; namely, the assumption that through a point outside a straight line there passes *more* than one-parallel line.

Now such a process is *structurally inherent in all human knowledge.* More than this, it is a unique characteristic of the structure of human knowledge. We can always do this. If we pass to higher orders of abstractions, situations seemingly 'insoluble', 'matters of fact', quite often become problems of *preference.* This problem is of extreme semantic importance, and of indefinitely extended consequences for all science, psychiatry, and education in particular.

The examples I have given show a most astonishing semantic situation; namely, that one question can sometimes be answered 'yes' or 'no', 'true' or 'false', depending on the order of abstractions the answerer is considering. The above facts alter considerably the former supposedly sharply defined fields of 'yes' and 'no', 'true' and 'false', and, in general, of all multiordinal terms. Many problems of 'fact' on one level of abstraction become problems of 'preference' on another, thereby helping to diminish the semantic field of disagreement.

It is interesting to throw some light on the problem of 'preference'. Which statement or attitude is preferable? The one claiming that Lobatchevski *dropped* a postulate, or the one claiming that Lobatchevski *introduced* a new postulate? Both are 'facts', but on different levels, or of different orders. The *dropping* appears as an historical fact; the *introducing* as a psycho-logical fact *inherent* in the structure of human knowledge. The preference is fairly indicated; the psycho-logical fact is of the utmost generality (as all psycho-logical facts are) and, therefore, more useful, since it applies to all human endeavours and not merely to what a certain mathematician did under certain circumstances.

Section C. Confusion of higher orders of abstractions.

We have already seen that Fido's power of abstracting stops somewhere. If we are finalists of any kind, we also assume that *our* power of abstracting stops somewhere. In some such way the finalistic, dogmatic and absolutistic semantic attitudes are built.

If, however, by the aid of the Structural Differential we train the *s.r* of our children in $\bar{A}$ non-identity and the inherent stratification of human knowledge and power of abstracting, we *facilitate* the passing to higher order abstractions and establish *flexible s.r* of *full conditionality* which are unique for Smith and of great preventive and therapeutic value. We thus build up 'human mind' for efficiency and sanity, by eliminating the factors of semantic blockages, while, by engaging the activity of the higher nerve centres, we diminish the vicious overflow of nervous energy upon the lower nerve centres, which, if allowed, must, of necessity, make itself manifest in arrested or regressive symptoms.

The above issues are of serious semantic importance in our daily lives and in sanity. All semantic disturbances involve evaluation, doctrines, creeds, speculations . , and vice versa. Under circumstances such as described above, which appear inherent with us, it is dangerous not to have means to see one's way clear in the maze of verbal difficulties with all their dangerous and ever-present semantic components.

By disregarding the orders of abstractions, we can manufacture any kind of verbal difficulties ; and, without the consciousness of abstracting, we all become nearly helpless and hopeless semantic victims of a primitive-made language and its underlying structural metaphysics. Yet the way out is simple ; non-identity leads to 'consciousness of abstracting' and gives us a new working sense for *values,* new *s.r*, to guide us in the verbal labyrinth.

Outside of 'objectification', which is defined as the evaluation of higher order abstractions as lower ; namely, words, memories . , as objects, experiences, feelings . , the most usual identification of different *higher order* abstractions appears as the confusion of inferences and inferential terms with descriptions and descriptive terms.

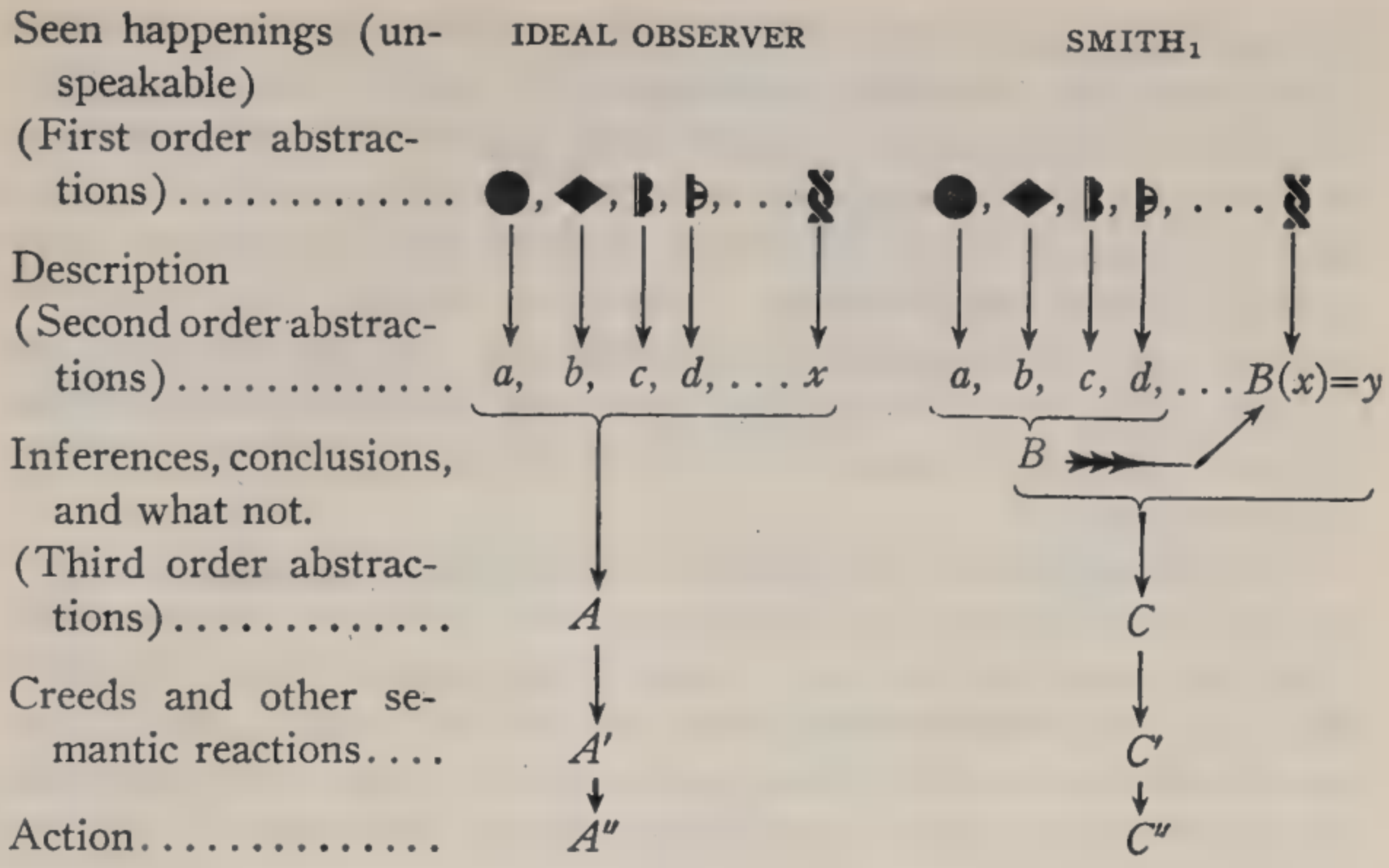
Obviously, if we consider a description as of the *n*th order, then an inference from such a description (or others) should be considered as an abstraction of a higher order $(n+1)$. Before we make a decision, we usually make a more or less hasty survey of happenings, this survey establishing a foundation for our judgements, which become the basis of our action. This statement is fairly general, as the components

of it can be found by analysis practically everywhere. Our problem is to analyse the general case. Let us follow up roughly the process.

We assume, for instance, an hypothetical case of an ideal observer who observes correctly and gives an impersonal, unbiased account of what he has observed. Let us assume that the happenings he has observed appeared as: ●, ◆, ▮, ▶, . . . , and then a new happening ▚ occurred. At this level of *observation,* no speaking can be done, and, therefore, I use various fanciful symbols, and not words. The observer then gives a *description* of the above happenings, let us say *a*, *b*, *c*, *d*, . . . , *x*; then he makes an inference from these descriptions and reaches a conclusion or forms a judgement *A* about these facts. We assume that facts unknown to him, which always exist, are not important in this case. Let us assume, also, that his conclusion seems correct and that the action *A″* which this conclusion motivates is appropriate. Obviously, we deal with at least three different levels of abstractions: the seen, experienced . , lower order abstractions (un-speakable) ; then the descriptive level, and, finally, the inferential levels.

Let us assume now another individual, $Smith_1$, ignorant of structure or the orders of abstractions, of consciousness of abstracting, of *s.r.*; a politician or a preacher, let us say, a person who habitually identifies, confuses his orders, uses inferential language for descriptions, and rather makes a business out of it. Let us assume that $Smith_1$ observes the 'same happenings'. He would witness the happenings ●, ◆, ▮, ▶, . . . , and the happening ▚ would appear new to him. The happenings ●, ◆, ▮, ▶, . . . , he would describe in the form *a*, *b*, *c*, *d*, . . . , from which fewer descriptions he *would form a judgement, reach a conclusion, B*; which means that he would pass to another order of abstractions. When the new happening ▚ occurs, he handles it with an already formed opinion *B*, and so his description of the happening ▚ is *coloured* by his older *s.r* and no longer the *x* of the ideal observer, but $B(x) = y$. His description of 'facts' would *not* appear as the *a*, *b*, *c*, *d*, . . . , *x*, of the ideal observer but *a*, *b*, *c*, *d*, . . . , $B(x) = y$. Next he would abstract on a higher level, form a new judgement, about 'facts' *a*, *b*, *c*, *d*, . . . , $B(x) = y$, let us say, *C*. We see how the semantic error was produced. The happenings appeared the 'same', yet the unconscious identification of levels brought finally an entirely different conclusion to motivate a quite different action, *C″*.

A diagram will make this structurally clearer, as it is very difficult to explain this by words alone. On the Structural Differential it is shown without difficulty.



Let us illustrate the foregoing with two clinical examples. In one case, a young boy persistently did not get up in the morning. In another case, a boy persistently took money from his mother's pocketbook. In both cases, the actions were undesirable. In both cases, the parents unconsciously identified the levels, x was identified with $B(x)$, and confused their orders of abstractions. In the first case, they *concluded* that the boy was *lazy;* in the second, that the boy was a *thief*. The parents, through semantic identification, read these inferences into every new 'description' of forthcoming facts, so that the parents' new 'facts' became more and more semantically distorted and coloured in evaluation, and their actions more and more detrimental to all concerned. The general conditions in both families became continually worse, until the reading of inferences into descriptions by the ignorant parents produced a semantic background in the boys of driving them to murderous intents.

A psychiatrist dealt with the problem as shown in the diagram of the ideal observer. The net result was that the one boy was not 'lazy', nor the other a 'thief', but that both were ill. After medical attention, of which the first step was to clarify the symbolic semantic situation, though not in such a general way as given here, all went smoothly. Two families were saved from crime and wreck.

I may give another example out of a long list which it is unnecessary for our purpose to analyse, because as soon as the 'consciousness of abstracting' is acquired, the avoidance of these inherent semantic difficulties becomes automatic. In a common fallacy of *'Petitio Principii'*,

or 'Begging the Question' fallacy, we, by self-deceptive semantic evaluation, *assume the conclusion* to be proved. In other words, we confuse the orders of abstractions. Beside the wilful use of this fallacy by lawyers in courts to influence juries of low intelligence., a similar fallacy is widely committed in the reasonings of daily life and leads to many unnecessary semantic difficulties. Particularly vicious is the use of the so-called 'question-begging epithets'. We postulate the fact which we wish to prove, label it by another name, and then use the new higher order name in our premise. It represents clearly a confusion of orders of abstractions.

All such terms as 'un-patriotic', 'un-christian', 'un-american', 'pro-german' (during the World War), 'wet', 'dry'., fall into this group. It is probably no secret that a large part of the population of this world was swayed by such methods during the war. In times of peace, large countries are continually swayed by such use of terms which play upon the pathological *s.r* of the population, thereby facilitating the 'putting over' of different propagandas. Similar procedures lead to many semantic difficulties in daily life. It is easy to see that the difficulty is general; namely, 'the confusion of orders of abstractions'. The antidote is equally general, and is found in the elimination of the 'is' of identity, resulting in the 'consciousness of abstracting'. It should be noticed that these pathological reactions have long been known, and that they are extremely general. We are told about them in schools under the name of 'logical fallacies', disregarding their semantic character, and so it is practically impossible to eliminate them or to apply the wisdom we are taught. It is not difficult to see why this should occur. In the older days, all the 'wisdom' was taught to us by purely 'intellectual', 'verbal', classical *A* and *el* methods. We had no simple psycho*physiological* method of *complete* generality, which could be taught in a *non-el* way affecting *all* nerve centres. It is known how difficult it is to 'change human nature', which simply means that the older verbal educational methods could not properly affect the lower centres. It seems that the first step in developing a method to accomplish these ends is to use the Structural Differential, without which it is practically impossible to teach 'silence on the objective level' and 'delayed action' and to train through *all* centres in non-identity, 'stratification', natural order, and so in appropriate *s.r*. It appears that now, to begin with, we have acquired a workable and simple psychophysiological method for changing identification into visualization, and, in general, for the prevention or elimination of identification or confusion of orders of abstractions. We have now discovered a mechanism which involves and deals directly with the reactions of the

lower centres, 'senses', affects, 'emotions',. The older, difficult 'change in human nature' becomes an easily accomplished fact in a structural, $\bar{A}$ semantic education. 'Human nature' can best be described, perhaps, as a complex of *s.r*, which *can* be educated and 'changed' to a large extent.

It seems unnecessary to enlarge further upon this subject. Every attentive reader can supply endless examples of this kind of semantic disturbances from his own observation or experience. Naturally, the generality, simplicity, and *physiological* character of the method proposed in this work become powerful assets, and instruction in the $\bar{A}$ methods can easily be given to, or acquired by, everybody. It can be taught in homes and schools. It gives a preventive psychophysiological method of training the *s.r* in the millions and millions of cases in which human life becomes wrecked through the lack of a *working structural educational theory* concerned with these reactions. But it is not enough to preach these 'platitudes'; they must be practised as well. If the parents and the boys mentioned above had been trained as children with the Structural Differential, it would have been an impossibility for the situation to have become so acute.

Let us follow our *daily experiences* by the aid of the Structural Differential. We find ourselves on at least five levels. The first represents the un-speakable event, or the scientific object, or the unseen physico-chemical processes on the sub-microscopic levels which constitute stimuli registered by our nervous system as objects. The second consists of the external, objective, also un-speakable, levels on which we see with our eyes,. On this level, we could make a moving picture, including actions., (writing a book is also behaviour). The third level represents the equally un-speakable psycho-logical 'pictures' and *s.r*. On the fourth level of abstractions we describe verbally our facts, that humans (*a*) eat, sleep.; (*b*) cheat, murder.; (*c*) moralize, philosophize, legislate.; (*d*) scientize, mathematize,. Finally, in the present context, our inferences belong to the fifth level.

Unfortunately, we usually abstract facts (*a*), identify the levels, and form a conclusion 'man is an animal',. From this *conclusion* we confuse the levels again and colour the description of the facts (*b*), (*c*), (*d*).; jump again to higher levels and build conclusions from descriptions (*a*) and from *distorted,* coloured descriptions (*b*), (*c*), (*d*), and so obtain the prevailing doctrines in all fields. These again lead us, in the field of action, to the mess we all find ourselves in. In this dervish dance between the levels we entirely *disregarded uncoloured facts* (*d*).

The ideal observer would observe *all* forms of human behaviour at a given date, *not leaving out facts* (*d*); then, without confusing his

levels, and also without confusing descriptions with inferences, he would reach his higher order of abstractions properly, with very different resultant doctrines, which would produce entirely different semantic evaluation, and motivate equally different action.

We may understand now why we must constantly revise our doctrines, for the above analysis throws a considerable light on the fact that scientists need training with the Differential as much as other mortals (the author included). History shows that they have not officially checked themselves up sufficiently to become aware of this fatal habit of confusion of orders of abstractions through identification.

It might appear, at first glance, that all that has been said here is simple and easy. On the contrary, it is *not* for the grown-ups; it is easy only for children and the young. In all my studies and experimenting I have found that, for the reasons already given, the use of the Differential appears essential, and that it requires a long while and training to accomplish new semantic results. As a rule, unless they are very unhappy, people try to trust their 'understanding', and dislike to train repeatedly with the Differential. For some reason or other, they usually forget that they cannot acquire structural familiarity with, or reflex-reactions in, spelling, or typewriting, or driving a car., *by verbal means alone*. Similar considerations apply in this case. Without the actual training with the Differential, certainly the best results cannot be expected.

To gain the full benefit involves the uprooting of old habits, taboos, 'philosophies', and private doctrines, the worst being the structure of our primitive *A* language with the 'is' of identity, all of which are deeply rooted and work unconsciously. Only the semantic training with the Differential in *non-identity* can affect the 'habitual' and the 'unconscious'. Rationalization, lip-service to the 'understanding' of it, will be of no use whatsoever. Persistent training seems the only way to acquire this *special structural sense for proper evaluation,* and the habit of *feeling* when identification, or the confusion of orders of abstractions becomes particularly dangerous. This feeling, as it involves most important factors of evaluation, is difficult to acquire, as difficult, perhaps, as reflex-learning to spell or to typewrite. But, when acquired, it makes us aware of the continuous, necessary utilization of many levels of abstractions, which becomes dangerous only when we identify them or are *not conscious* of this fact. We can then utilize the different orders of abstractions *consciously,* without identification, and thus keep out of danger. Most of the important *terms* appear as multiordinal, and, although they *belong* to verbal levels, they *apply* often to all levels, an

important structural fact impossible to avoid, and one which makes this special semantic sense uniquely necessary to acquire.

It seems unnecessary to repeat that everything that has been said above applies in the fullest extent to our ethical, social, political, economic, and international relations. Before any sanity can be brought into the analysis of these relations, before they can be rationally analysed, the investigators would have to be trained to observe correctly and to avoid verbal structural pitfalls. For the lack of such semantic training and re-education, the 'time-honoured' 'Fido' debates involving the 'is' of identity, continue on all sides, and lead to naught else but a waste of 'time' and effort.

I say waste of 'time', simply because there seems no end to the paradoxes which, with a little ingenuity, we can build up when we begin to gamble with confusion of orders of abstractions and disregard multi-ordinality. Any doctrine, no matter how structurally true or beneficial, can be defeated, confused, or delayed, by the use of such methods. These problems appear of crucial semantic importance, because our lives are lived in a *permanent* structural interplay between different orders of abstractions. All our achievements depend upon this interplay, yet the most acute and painful dangers also have their sources in the non-realization of this dervish dance between different orders of abstractions.

Since we cannot evade the passing from level to level, or the use of multiordinal terms, our wisdom should consist only in not abusing these semantic conditions of human life. As we must do that, let us do it, but let us not identify the orders, and thus let us evade the dangers. Consciousness of abstracting gives us the complete *psychophysiological* solution of this complex situation, as it allows us to have the psychological benefits and to avoid the dangers by the use of *physiological* means.

In conclusion, I must stress once more the importance of the structure of the language in which we analyse any given problem. In the $\bar{A}$-system I am proposing, the term *order* is accepted as one of its very foundations. In 1933, we know that as words *are not* the things spoken about . , structure, and structure alone, becomes the only possible content of knowledge, and the search for structure, the only possible aim of science. If we try to define structure, we can do so in terms of relations and multi-dimensional order. The recent advances of science show, beyond doubt, that the day will come when all science will be formulated in terms of structure and, therefore, of physics, and physics formulated as a form of multi-dimensional geometry, based on multi-dimensional order, giving us, ultimately, multiordinal structure.

The application of the term *order,* which involves physiological, as well as semantic, mechanisms of evaluation, to the analysis of human behaviour, has led me to the present $\bar{A}$-system and the investigation of the structure of language. The discovery that some of the most important terms we use appear multiordinal, a character *concealed* by the 'is' of identity, has disclosed to us a most vital and inherent psychological mechanism, responsible in humans for many most desirable, many undesirable, and many morbid human characteristics. It disclosed, also, the psycho-logical *structure* of these characteristics, and so we have obtained *physiological* means by which to enhance the development of desirable characteristics, and to prevent or transform the others.

Further analysis has disclosed a natural survival order in evaluation: the event first, the object next; the object first, the label next; description first, inferences next., in inherent importance. We have also found that the majority of human difficulties, 'mental' ills included, involve semantic disturbances and exhibit, *not* the natural survival order, but the identification of different orders, resulting in a reversed (pathological) order.

It is impossible in this book to review the data of psychiatry from the $\bar{A}$ point of view, as this would require a separate large volume, which, I hope, will be written some day; but any one can verify the statements made above by himself from clinical literature, and also by analysing his own or other persons' life-difficulties, quarrels, disagreements., which generally involve quite unnecessary sufferings. Psychotherapeutic literature shows abundantly that the success of the physicians depends mostly on reversing the pathological reversed order in a given field, and so restoring the natural order in the *s.r.* It is easily verified that, in most cases, when 'mental' illness originated through different life experiences, these would have affected very little, if at all, a child or an adult who was conscious of abstracting, and whose nervous processes and corresponding semantic states followed the natural order.

With the aid of the Structural Differential and a $\bar{A}$ language of new structure, it is easy to train the *s.r* of an infant, a child, or a young person, and possible, although much more difficult, to train a grown-up in the natural order. Such a training becomes a potent preventive structural *physiological* method, as it eliminates the psycho-logical states of identification or reversed order, both of which represent the raw semantic material out of which future nervous disorders are produced.

The *non-el* term 'order' is equally applicable in life and science; gives us, in 1933, the simplest structural common base, and allows us to

attempt the formulation of a science of man, which ultimately becomes a theory of sanity, a consequence of a *non-aristotelian system.*

It should be recalled that order is accepted in the present system as undefined and fundamental; yet its use is easily explained by the aid of the term 'between', and can be shown and applied in reference to empirical structures.

If we can formulate a method which, through the application of a psychophysiological term such as *order,* and a simple device such as the training of *s.r* in the natural survival order, or reversing the pathological reversed order, includes the mechanism of non-identity and one of the most important human nervous functionings, such a method, because of its structural simplicity and physiological character, may be expected to prove very workable. I desire to stress most emphatically the very important *general, impersonal, preventive, semantic, reflex-character* of such a method.

In actual life, we deal, for the most part, with persons who are 'mentally' or nervously disturbed in different degrees. We could, eventually, divide them for our purpose into two groups: (1) those who do not want to improve or get well, but who somehow like their fictitious worlds and the maladjustments connected with them; (2) those who genuinely want to get over their difficulties.

In general, it is extremely difficult or impossible to achieve anything at all with the first group. The second group is greatly helped if we give them means to work by themselves at their problems. Very often it is most effective to explain to them this simple 'natural order', 'identification', and 'reversed order' mechanism, the multiordinality of terms., and so give them a *definite psychophysiological symptom* to struggle against. These symptoms of identification or reversed order, in their generality and structural neurological fundamentality, underlie the process of formation of practically all known semantic difficulties of evaluation.

The reader should not assume that it is always possible to eliminate identification and so achieve this coveted natural order, or the reversal of the reversed pathological order; but, whenever this *is* possible, the person is relieved in a great many psycho-logical fields. The simplicity and generality, the physiological and structural character of this method seems its main recommendation, particularly as a preventive measure or semantic training for sanity. The training is a laborious process, requiring great persistence; but, to my knowledge, very few trainings are easy, and, perhaps, none leads to more important results than does this one.

CHAPTER XXVIII

ON THE MECHANISM OF IDENTIFICATION AND VISUALIZATION

"Did you say 'pig,' or 'fig'?" said the Cat.

"I said 'pig'," replied Alice; "and I wish you wouldn't keep appearing and vanishing so suddenly: you make one quite giddy!"

"All right," said the Cat; and this time it vanished quite slowly, beginning with the end of the tail, and ending with the grin, which remained some time after the rest of it had gone.

"Well! I've often seen a cat without a grin," thought Alice; "but a grin without a cat! It's the most curious thing I ever saw in all my life!"*

LEWIS CARROLL

The significance of the paradoxical phase is not limited to pathological states such as those previously observed, and it is highly probable that it plays an important part in normal men too, who often are apt to be much more influenced by words than by the actual facts of the surrounding reality. (394)

I. P. PAVLOV

In the case of an imbecile, repetition without comprehension, psittacism, may prevail; the rôle of visual impressions is null or nearly so among the illiterate; the deaf from birth who have learned to speak have no auditory impressions to intervene. But, normally, it is feelings and ideas that appear in action, in the form of language. (411)

HENRI PIÉRON

The specific neurones necessary for sensation are also necessary for the associative reawakening of that sensation, which is called the image—a dynamic process and not a photographic negative resting miraculously in the nervous substance, where some subtle spirit might go to consult it. (411)

HENRI PIÉRON

It is none the less true that certain cultivated persons can use visual images, and can even use these images in preference to others. (411)

HENRI PIÉRON

Objectification and visualization are usually not differentiated. The first represents a very undesirable semantic process, whereas the second, visualization, represents one of the most beneficial and efficient forms of human 'thought'. From a $\bar{A}$ point of view, such a lack of differentiation between the two reactions appears as a very serious problem, requiring an analysis of the respective mechanisms.

To visualize, we must have such forms of representation as lend themselves to visualization; otherwise, we must fail. The A-system, which could not adequately handle asymmetrical relations, and could not be built explicitly on structure, necessarily involves identification. In the A period, we were able to visualize objects and a few objective situations, but all the higher abstractions were, in principle, inaccessible to visualization, making scientific theories needlessly difficult. A $\bar{A}$-system, free from identification, must be based explicitly on *structure* on all levels (structure defined in terms of relations and ultimately multi-dimensional

Alice in Wonderland.

order), which can be easily visualized. It should be recalled that structure, relations, and multi-dimensional order supply us with a language which completely bridges daily-life experiences with all science, leading toward a *general theory of values*. Mathematics and mathematical physics then become the representatives and the foundation of all science; and in the human field a general theory of values will lead to adjustment or sanity and will some day include ethics, economics,.

For these reasons, the Structural Differential is uniquely useful, as, at a glance, it conveys to the eye structural differences between the world of the animal, the primitive man, and the infant, which, no matter how complex, is extremely simple in comparison with the world of the 'civilized' adult. The first involves a *one-valued orientation* which, if applied to the ∞-valued facts of life, gives extremely inadequate, wasteful, and ultimately painful adjustment, where only the few strongest survive. The second involves ∞-valued orientation, similar in structure to the actual, empirical, ∞-valued facts of life, allowing a one-to-one adjustment in evaluation with the facts in each individual case, and producing a semantic flexibility., necessary for adjustment. This flexibility is known to be the foundation for balanced semantic states, 'higher intelligence',.

Visualization requires a definite elimination, through differentiation, of harmful identification, which, as usual, is based on incorrect evaluation of structural issues. Thus, we have had endless, bitter, and futile arguments as to whether or not the 'mechanistic' point of view of the world and ourselves is legitimate, adequate,. The average person, as well as the majority of 'philosophers', identifies '*me*chanistic' with '*ma*chinistic'. Roughly, mechanics is a name for a science which deals with dynamic manifestations on all levels; thus, we have macroscopic classical mechanics, colloidal mechanics now being formulated, and the sub-microscopic quantum mechanics already being well-developed disciplines. In the rough, 'machine' is a label applied to a man-made apparatus for the application or transformation of power. But even machines differ greatly; thus, a dynamo is entirely different in principle, in theory, and in applications from a lathe or an automobile.

If we ask: 'Is the *ma*chinistic point of view of the world justified?', the answer is simple and undeniable; namely, that this point of view is grossly inadequate and should be entirely abandoned. But it is not so with the *me*chanistic point of view, understood in its modern sense and including the quantum mechanics point of view, which is entirely *structural*. In 1933, we know positively that even the gross macroscopic physico-chemical characteristics of everything we are dealing with depend on the sub-microscopic *structure* (see Part X). The details are

not yet fully known, but the principles are firmly established. With a $\bar{A}$ understanding and evaluation of the unique importance of structure as the only possible content of 'knowledge', these 'firmly established' principles become *'irreversibly established'*. We may go further and say that the quantum mechanics point of view becomes the first structurally correct point of view and, as such, should be accepted fully in any sane orientation. If we stop identification, then we will differentiate between some simple facts. For instance, we will understand that any semantic state, reaction, or process has its corresponding sub-microscopic, structural, colloidal, and ultimately quantum mechanical processes going on in the nervous system; however, the *s.r*, or feelings of pain or pleasure., *are not* the sub-microscopic processes. These belong to different levels, but with ∞-valued semantics we can establish in principle a one-to-one correspondence between them. Thus, when we differentiate adequately, the older *mach*inistic objections disappear entirely; and, in its proper field, for structural reasons, we must preserve the *mechanistic,* and entirely abandon the too crude *mach*inistic attitudes. The *mechanistic* (1933) attitude is based on *structure* and so is indispensable for visualization; and *training in visualization automatically abolishes objectification,* which represents an important special case of all identification. From the point of view of a $\bar{A}$-system, adjustment and sanity in humans depend, to a large extent, on their 'understanding', which is entirely structural in character; therefore, we must accept a *mechanistic* (1933) *attitude,* which, in the meantime, can be visualized.

The finding of structural means of representation facilitates *visualization,* imagining, picturing,. In the adjustment trend we start with lower nerve-impressions, 'senses', 'feelings'., lower abstractions, and these are abstracted again by the higher centres. The higher centres produce the 'very abstract' theories, which cannot be visualized for a while. The lower centres, which are involved in visualization, can deal only with structures which can be 'concretely pictured'. So we always try to invent mechanistic or geometrical theories, such as can be handled by the lower centres.

Individual 'experiences', supplied by the lower centres of different individuals, do not blend directly. They are blended in the higher centres. In them manifold experiences, whether individual or accumulated by the race (time-binding), are abstracted further, integrated, and summarized. Once this has been accomplished, structural means are sought *and discovered* to translate these higher abstractions into lower, the only ones with which the lower centres can deal. Then we can 'visualize' our theories, and the higher centres not only influence the lower centres, but

the lower centres have appropriate means by which to co-operate with the higher centres in their new *non-el* quests.

The lack of explicitly structural forms of representations is responsible, also, for the difficulties which arise when the higher order abstractions are translated into the reflex-reactions of the lower centres, which can deal with 'intuitions', 'orientations', 'visualization',. The so-called 'geniuses' have a very subtle nervous system in which the translation of higher order abstractions into lower and vice versa is easily accomplished. From the point of view of forms or representations, we can have two issues: (1) we may have *el* forms of representations which are not based on structure, visualization., and cannot efficiently affect the activities of the lower centres; (2) we may have a *non-el* system based on structure, visualization., which can be translated simply, easily, and efficiently into the terms of the lower centres. These problems are of educational importance and should be worked out more fully.

In my experience with grown-ups who have had only a *short* contact with my work, I find, in many cases, that, although they may have even given their complete verbal approval of the main point of the system, yet, invariably, in practice, the full application is lacking. Obviously, the semantic importance of the present findings is not in the verbal approval alone, when that approval is not applied, but in the consistent and permanent instinctive acquisition of the new semantic attitude which involves a complete elimination of identification, allness, elementalism,.

We can teach any one to repeat verbally, by heart, instructions for operating an automobile, a piano, or a typewriter; but no one could operate them satisfactorily by reflex-action after such verbal training alone. To operate effectively and skilfully any structural complex, we must become intimately familiar with its structural working through actual reflex-training, and only then can we expect the best results. In my experience, this is true with language, and, without the *visual* Structural Differential on which we can point our finger to the objective level and urge silence., such basic semantic *reflex-training* cannot properly be given.

If we ask a man: 'Do you know how to drive a car?', and he answers 'Yes', we assume that he has acquired the proper *reflexes*. If he answers 'No, but I know *about* it', he means that he has *not* acquired the proper reflexes, but that his 'knowledge' is on purely verbal levels, non-effective in application on *non-verbal* reflex-levels. This applies fully to *s.r*; we may 'know' *about* them, but we may never apply successfully what we supposedly 'know'. To 'know' represents a multiordinal process which involves equally the activities of the lower nerve centres and of the higher. In our *el* systems we had no such distinction, and so we

confused them. The older 'knowledge', when presented in *el* language, could not have been absorbed easily by the *non-el* organisms-as-a-whole. As the main task, at present, is to unlearn the older *s.r*, the new reactions need a persistent training, particularly by the grown-ups. The *non-el*, $\bar{A}$ language and method prove to have psychophysiological importance.

Although the neurological mechanism underlying identification, objectification, visualization., is not well known (1933), neurology gives us evidence that in these states, as well as in delusions and hallucinations, the actual lower nerve centres are somehow engaged. We may assume that different 'resistances', 'blockages'., in some parts of the nervous system make the passage of nervous impulses more difficult, and it seems reasonable to suppose that, in such cases, the paths travelled by the nervous currents are different.

In Fig. 1, an hypothetical and over-simplified scheme of the different types of distribution of nervous currents, as is known functionally, is suggested. The ordering is not anatomical but functional in terms of degrees of intensity. In this scheme, we may consider that the nervous impulse (A) reaches the lower nerve centres, the brain-stem and the thalamus, passes through the sub-cortical layers and the cortex, continuously being transformed. Finally, in returning, it may take either the beneficial and adaptive semantic form of visualization (V), free from identification and semantic disturbances, or may involve identification, with semantic disturbances, such as objectifications of different orders (O), delusions (D), illusions (I), or, finally hallucinations (H).

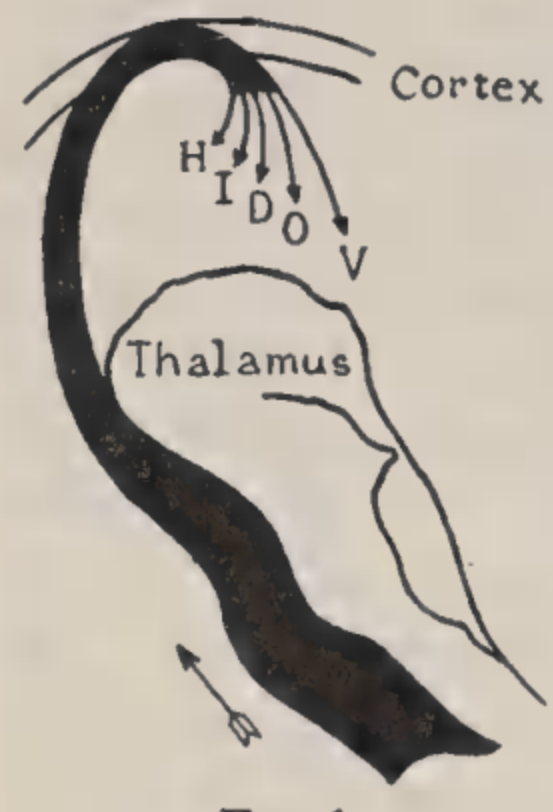


FIG. 1

Identification, or confusion of orders of abstractions, consists of erroneous evaluation: that which is going on inside of our skin has objective existence outside of our skins; the ascribing of external objectivity to words; the identification in value of 'memories of experiences' with experience; the identification of our *s.r* and states with words; the identification of inferences with descriptions,. Identification is greatly facilitated, if not actually induced, by the *A* structure of language in which we have *one name* for at least *four* entirely different entities. Thus, the *A* 'apple' (without subscripts and date) is used as a label for the physico-chemical process; for an object, say, 'apple$_{1,\text{Feb.23,1933}}$'; for a 'mental' picture on the un-speakable semantic level, and for the verbal definition. Under such linguistic conditions, it is practically impossible, *without special training,*

not to identify the *four* entirely different abstractions into *one*., with all the following sinister consequences.

Delusions represent incorrect notions and inappropriate *s.r* formed, not by insufficient knowledge or 'logic', but by affective pressure in a definite evaluational direction; as, for instance, delusions of grandeur; delusions of persecution; delusions of 'sin'; delusions of reference,.

Illusions appear more like real perceptions, but pathologically changed. For instance, anything may be semantically coloured or interpreted, or evaluated as an offense, or a threat, or a promise,.

Hallucinations consist of 'perceptions', with all their vividness, but *without* any external stimuli. Patients hear voices; see visions; feel pricks or burnings., when there is nothing to hear, or see, or to be pricked by.

In *visualization,* identification does not occur; orders of abstractions are not confused; semantic disturbances do not appear; the *evaluation is correct;* a 'picture' is evaluated as a picture and not as the events,. In other words, because of the consciousness of abstracting, the natural order of evaluation is preserved. But once, through identification, this natural order is reversed, it marks a pathological condition more or less morbid, and often of a non-adaptive character.

Identification represents, in affective tension, the mildest semantic disturbance, consisting of an error in meanings and evaluation. Objects are evaluated as events; 'ideas', or *higher* order abstractions, are evaluated as objects; as experience; as the un-speakable semantic states or reactions; otherwise, as *lower* order abstractions. The confusion in the field of higher order abstractions follows a similar rule. Inferences obviously represent higher order abstractions than descriptions; so, when they are not differentiated, higher order abstractions are again identified with the lower. We all know from daily-life experience the fantastic amount of suffering we can, and do, actually produce for ourselves and others with such identifications.

In delusions, a similar but more intense identification occurs, resulting in erroneous semantic evaluation; wishes, feelings, and other semantic states inside of our skins are projected into the external world, giving delusionally strong objective evaluation.

In illusions, we also ascribe to, or identify our complex semantic states with, different perceptions and evaluate our higher order abstractions as lower.

In hallucinations, this process of reversing the natural order comes to a culminating point: higher order abstractions are translated into, and have the full vividness and 'reality' of, lower order abstractions.

We see that the pathological processes of 'mental' illnesses involve identification as a generalized symptom; which means the reversal, in different degrees, of the natural order of evaluation based on the intensified confusion of orders of abstractions. The more intense this process of reversal becomes, the more non-adaptive and morbid the manifestations. It should be noticed that this analysis becomes a necessity once we decide to accept a *non-el* language. This analysis is far from exhaustive, but an analysis in new *non-el*, structurally correct terms, throws a new light on old problems.

Hallucinations which result from 'physical' illness do not represent a permanent danger, but when a patient seems 'physically' well, and his confusions of orders of abstractions, delusions, illusions, and hallucinations become completely 'rationalized', then these are unmistakable signs of serious 'mental' illness, suggesting sub-microscopic colloidal lesions. Now this 'rationalization' represents nothing else but a nervous disturbance and involves *identification* somewhere. In 'physical' ills the nervous system may be disturbed, but the illness does not usually originate in nervous disturbances, and so, as such, is not dangerous.

The distinction between visualization and objectification based on a $\bar{A}$-system seems new; the difference is subtle, but when it is formulated we can discover a simple means whereby to control the situation. If we were to take a 'bone' made of papier-maché and smear it with fat or meat, Fido would, perhaps, *objectify* (identify) such a 'bone' from the smell and the form of the papier-maché with an edible one, and would fight for it. We do a similar sort of thing when we objectify. Religious wars, the 'holy inquisition', the persecution of science, which we are witnessing even at the present day in some countries and communities, are excellent examples.

We should notice that Fido was able to *trust* his natural, even 'objectified', instinct, for nature does not play such tricks on him, such as producing 'bones' of papier-maché. If nature did, dogs that objectify and persist in their liking for such 'food' would soon be eliminated. These particular objectifications would be dangerous and painful to those particular kinds of dogs with that particular nervous system, and would ultimately prove of no survival value. Thus identification, which represents an inappropriate evaluation, is harmful to all life, but is little noticed at present, because the main periods of the animal racial adjustment have been accomplished long ago. Experiments on flies show that the number of mutants which may be produced in a laboratory is large, but very few would survive outside of a laboratory. In unaided nature, these mutants probably occur, but seldom leave observable traces.[1]

However, even today, as Pavlov has shown in his laboratories, we can impose, by an interplay of a four-dimensional order of stimuli, such conditions upon animals for which their nervous survival structure was not naturally adapted, and so induce nervous pathological states. Wrong evaluation is, indeed, harmful to all life and accounts for such rigid survival laws in nature, which science teaches humans how to make more flexible. Practically word for word, this applies to ourselves. We are constantly producing more and more complex conditions of life, man-made, man-invented, and deceptive for the non-prepared. These new conditions are usually due to the application of the work of some genius, and the nervous system and *s.r* of most of us are not prepared for such eventualities. In spite of inventions and discoveries of science, which are *human* achievements, we still preserve *animalistic* systems and doctrines which shape our *s.r*. Hence, life becomes more strained and increasingly more unhappy, thereby multiplying the number of nervous break-downs.

It is known that not all people are able to visualize equally well. In the older days this fact was taken for granted, and did not suggest further analysis. Under present conditions in many human beings and also in animals, as shown in the experiments of Pavlov, the visual stimuli are physiologically weaker than the auditory ones; in man, however, the visual stimuli should be physiologically stronger than the auditory. This difference does not affect the *general* mechanism of the cyclic nerve currents and orders of abstractions. In the auditory type the main returning currents are deviated into different paths. The division between 'visual' and 'auditory' types is not sharp. In life we deal mainly with individuals who have no more than a special inclination for one or the other types of reaction.

In the case of 'mental' processes, human adjustment has to be managed on higher, more numerous, and more complex levels. Obviously, then, the auditory types are more enmeshed by words, further removed from life than the visual ones, and so cannot be equally well adjusted. This fact should not be neglected, and on the human levels we should have educational methods to train in visualization, which automatically eliminates identification.

The auditory channels which connect us with the external world are much less subtle and effective than the visual ones. The eye is not merely a 'sense-organ'. Embryology shows that the eye is a part of the brain itself, and what is called the 'optic nerve' must be considered not a nerve but as a genuine nervous tract. This fact, of course, would assign to the eye a special semantic importance, not shared with other

'sense-organs' or receptors. We ought not to be surprised to find that the visual types are better adjusted to this world than the auditory types. In pathological states, such as identifications, delusions, illusions, and hallucinations, there seems to be involved a translation of *auditory* semantic stimuli into visual images. In these pathological cases the order of evaluation appears as label first and object next, while the adaptive order seems to require object first and label next,. There seems little doubt that visualization is very useful, and that identification is especially harmful. The *most effective means to transform the s.r of identification is found in visualization, which indicates its special semantic importance.*

The semantic *disturbance* of identification may have many sources, auditory included, but the only adaptive trend is in visualization, which involves in some way the optical neural structure. Some structural light is thrown on this subject when we realize that, physiologically, the eye is more closely related to the vegetative nervous system, which regulates our vital organs, than the ear is. In man the optic thalamus is greatly enlarged, so that the whole thalamus is often called the 'optic thalamus'. Actually, the thalamus has many functions, other than visual, and is connected with affective manifestations.

As most of our observations are accomplished by the aid of the eye, we should expect auditory types to be *poor observers,* and so racially, in the long run, not so well adjusted semantically. Observation shows that the auditory types often have infantile reactions—a serious handicap. From an adaptive point of view the 'normal', non-infantile, best-adjusted individual ought to be a visual type. Auditory types must also be further detached from actualities than the visual types, as auditory stimuli involve more inferences than descriptions, which is the opposite of the functioning of the visual types. If inferences, rather than descriptions, are involved, we naturally deal with higher abstractions first, and with the lower next, and so there is always a danger of the semantic confusion of orders of abstractions, which necessarily involves inappropriate evaluation, of which objectification is only a particular case.

Even to common sense it seems clear that there is a significant difference between 'knowing' this world by hearing and 'knowing' it by seeing. There is, likewise, a difference between the translation of higher abstractions into lower terms by the visual path, and the corresponding translation by the auditory path. In daily life we never say 'I hear' when we wish to convey that we understand; but we say 'I see'. When we say 'I hear', we usually wish to convey that we have heard some-

thing which we did not fully grasp or approve. The above relation is rather important, but has not been sufficiently analysed. The problems of introversion and extroversion are connected with it.

The relation between the problems of identification and the *number* of values *found* in the empirical world in connection with the *number* of values *ascribed,* or *assumed*., by our semantic processes, is most important.

The following analysis is, by necessity, one-sided, over-simplified., as a fuller analysis would require a separate volume. I consider many problems 'in principle' only; this allows me a briefer treatment necessary for my purpose, but it must be realized that our language and general semantics, which, *in practice,* we use unconsciously, are *extremely complex* and involve one-, two-, three-, and ∞-valued components, never, as yet, sharply differentiated nor formulated. Investigation shows that the ∞-valued semantics is the most general and includes the one-, two-., and few-valued semantics as particular cases. The one-valued semantics of literal identifications are found only among animals, primitive people, infants, and the 'mentally' ill, although more or less serious traces of some identification are found in practically all of us, because these are embodied in the structure of our language and prevent the acquisition of the ∞-valued systems necessary for sanity. For my purpose, it is enough to formulate the problems for the complete elimination of primitive identification, and then modern, ∞-valued, $\bar{A}$ semantics follow automatically. Under such conditions, I must concentrate on the vital problem of one-valued identification and treat the two-., and few-valued systems sketchily, 'in principle', although we must realize that the last systems have been made more flexible by the use of many ingenious verbal devices which I do not even mention in the present work.

Let me repeat that the attitudes, flexibility, or fixity., of our *s.r* depend to a large extent on the structure of language used, which involves also its appropriate general semantics. The 'logic' of our schooldays represents a composite affair, in the main *A*, and we call it by this last name. This 'logic' can be considered as a two-valued 'logic' because of the fundamental 'law of the excluded third', expressed as 'A is B or not B', by which a third possibility is excluded. But even the traditional 'logic' had to admit in its scheme what was called 'modality'; namely, some degrees of certainty or uncertainty with which a given statement is made. Lately, Łukasiewicz has shown that a three-valued 'logic' can be so formulated as to include modality. Later, he and Tarski generalized it to an *n*-valued 'logic'. When *n* tends toward infinity, this 'logic' becomes the 'logic' of probability. If these disciplines are made *non-el,*

we have what I call one-, two-, three- . , and ∞-valued *general semantics*. Theoretically, and in practice, we are interested mostly in the one-, two-, three-, few-valued, and ∞-valued general semantics. For my purpose, and for simplicity, I shall deal only with identification; that is, the primitive one-valued semantics, the influence of which is found in both the two- and three-valued semantics, *and may only be completely eliminated in an ∞-valued semantics.*

We live in a four-dimensional space-time manifold which, on all levels, consists of absolutely individual events, objects, situations, abstractions . , and we must conclude that structurally we live in an *indefinitely many-valued* or ∞-valued world, the possibilities of which follow in principle the laws of combinations of higher orders. The above statement represents a description of a structural observation about the empirical world, independent of our pleasure, and can be contradicted only by exhibiting empirically, actual 'identity' or 'absolute sameness'. , of different events, objects, or situations . , which exhibiting becomes an impossibility if we decide to investigate facts more fully.

Under such empirical conditions, for adjustment and so for sanity, we must have on semantic levels such theories, systems, methods . , which would allow us in a given case, under given conditions, at a given date . , to evaluate the individual happenings *uniquely;* or which would allow us to establish a one-to-one correspondence between the essentially ∞-valued facts of experience and our semantic states. It becomes obvious that this can be accomplished only if we have ∞-valued and *non-el* general semantics. We see that the two-, or three-valued, *el A* 'logic', 'psychology'. , and, in general, the *A*-system, being structurally different from the empirical world, will prevent, in principle, such an adjustment and, therefore, sanity.

Identification may be considered as the remains of pre-human, or primitive, or infantile, one-valued semantics, which establishes, or results from, semantic states, by which the essentially ∞-valued facts of experience are not differentiated or evaluated properly, and so the indefinitely many values of these facts are identified into a single value. Such identification is always structurally unjustified and dangerous, and may be the result of a great many factors, such as low development, ignorance, insufficient observation, 'wishful thinking', fears, pathological states of our nervous system, different semantic disturbances, 'mental' ills, infantilism in the grown-ups , . But among humans we cannot avoid training, through the mechanism of language and its structure, in some, most often unconscious, general semantics, and so a great deal depends

on what kind of semantics or methods of evaluation we impose on our children.

We should notice an important fact which is usually disregarded; namely, that a language, and often one word, involves a definite type of semantics. Thus, in primitive 'polysynthetic' languages, it is not a question of associations or superstitions; the mystic characteristics and the thing simply are not differentiated, but literally identified into one whole. Thus, we have one-valued semantics where the 'good' and 'evil spirits' actually participate in everything considered as a synthetic whole.[2]

A language of 'true' and 'false' involves two-valued semantics; the introduction of adverbs or their equivalents introduces modality and so three-valued semantics. The introduction of indefinitely many degrees between the 'true' and 'false' leads finally to ∞-valued semantics.

A diagram may help to make this clearer.

A, B, C., ∞-valued and different facts of experience, which, in a given case, have, by necessity, *indefinitely many, single, individual* values.

a, b, c., ∞-valued non-aristotelian orientation structurally *similar* to the empirical world which allows us, in a given case, to assign indefinitely many single, one-to-one corresponding values to the individual facts.

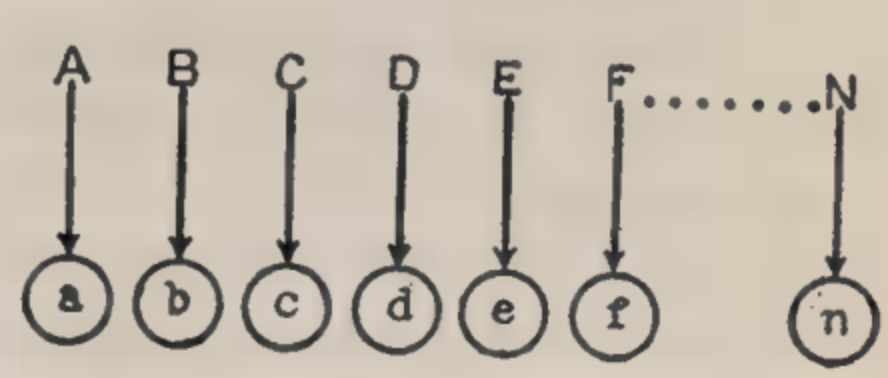


FIG. 2

A, B, C., ∞-valued and different facts of experience, which, in a given case, have, by necessity, *indefinitely many, single, individual* values.

Σ_1, Σ_2., two-, three-., and few-valued aristotelian orientation structurally non-similar to the empirical world, which compels us to ascribe two., or few values to the essentially indefinitely many-valued and different facts, resulting in identification of the many values into a few, which improper evaluation is projected on the facts.

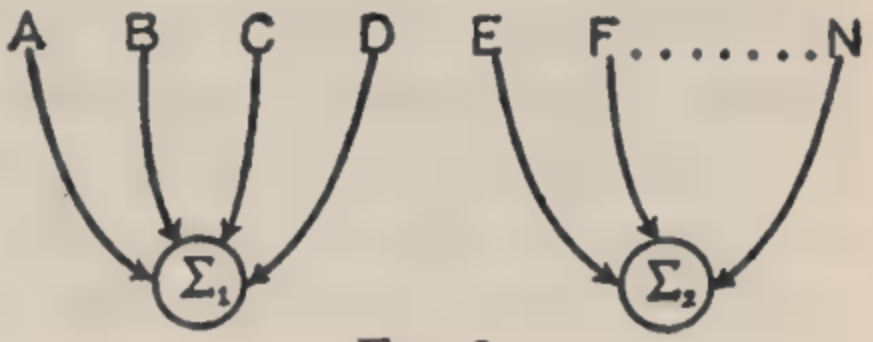


FIG. 3

A, B, C., ∞-valued and different facts of experience, which, in a given case, have, by necessity, *indefinitely many, single, individual* values.

Ω, one-valued, animal, primitive., orientation, structurally non-similar to the empirical world, which compels us to ascribe one value to the essentially indefinitely many-valued and different facts, resulting in *identification* of the many values into one, which improper evaluation is projected on the facts.

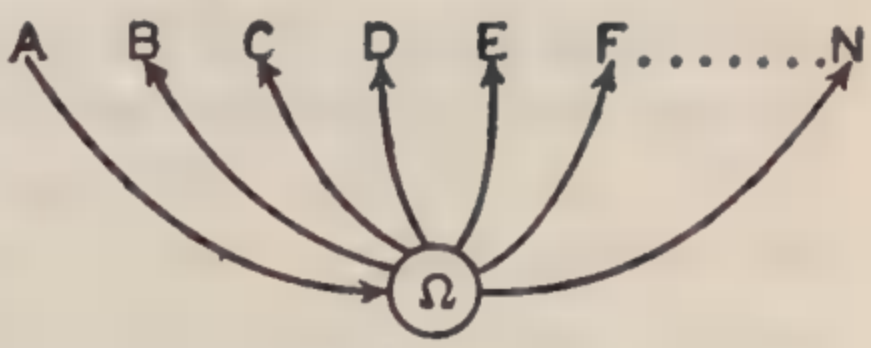


FIG. 4

In Fig. 2, the arrows Aa, Bb . , indicate the $\bar{A}$ one-to-one correspondence between the ∞-valued individual facts of life, A,B,C . , and the corresponding *s.r*, or a,b,c orientations . , which ascribe single values to the different facts, establishing a foundation for structurally correct *proper evaluation* which helps adjustment and so sanity.

In Fig. 3, the *A* two- . , few-valued orientation and type of correspondence is shown.

In Fig. 4, Ω indicates a single, say, proper evaluation of the *one* fact A. The arrows ΩB, ΩC, ΩD, ΩE, ΩF, . . . , ΩN indicate the *projection* of the one-valued semantic state, or orientation on the essentially unchanged ∞-valued facts A, B, C . , *distorting them*. In other words, the ∞-valued facts, through the identification of many values into one, and by pathological projection, have been given wrong evaluation, thereby preventing, *in principle,* adjustment and sanity, particularly for a civilized human 1933.

If we train our children in one-, two-, three-, and more generally few-valued *el, A* reactions based on corresponding languages, 'logics' . , the result must be that they will have great difficulty in adjusting themselves to a world of *non-el* ∞-valued facts, and that, even if they succeed, this would ultimately happen only after a great waste of efforts and unnecessary sufferings. If we approach the ∞-valued facts of life with one-, two-, or even few-valued semantic attitudes, we must identify some of the indefinitely many values into one, or a few values, and so approach the ∞-valued world with an orientation which *projects* ignorantly or pathologically our *restricted,* few-valued semantic evaluations on the ∞-valued individual facts of experience.

The above explanations apply in the fullest extent to the structure of language. The daily language, as well as our attitudes toward it, still reflects primitive structural *s.r* of the period before it was known that on the objective levels we deal *exclusively* with *∞-valued, four-dimensional processes.* The language in the *A*-system represents, *in principle,* what may be called a three-dimensional and one-, two- . , more generally few-valued linguistic system structurally non-similar to the ∞-valued, four-dimensional event-process conditions. Let us analyse, for instance, the *A* term 'apple'. This term represents, *in principle*, a name for a verbal, one-valued, and constant intensional definition, in which space-time relations do not enter. What are the structural facts of experience? The object which we call 'apple' represents a process which changes continually ; besides, every single apple that ever existed, or will exist, was an absolute individual, and different from any other objective 'apple'. In applying such a three-dimensional and one-valued language to essen-

tially ∞-valued processes, we only make proper evaluation, and so adjustment and sanity, very difficult.

Yet the structural adjustment is simple in a $\bar{A}$-system. The A 'apple' was a name for a verbal *intensional definition;* in a $\bar{A}$-system, we manufacture indefinitely many names for the indefinitely many objective and different 'apples' by subscripts, 'apple$_1$', 'apple$_2$', 'apple$_3$'., supplementing the subscript with the date; thus, in 'apple$_{1,\text{Feb.23,1933}}$', we gain the possibility of considering 'apple$_{n,t}$' as ∞-valued, and so, in a given case., we are enabled to have a single name which we could relate to single values of the objective, absolute individuals, and absolute individual stages of the process. Similarly with multiordinal terms. Before the multiordinality of terms was discovered and formulated by me in 1925, these terms were silently assumed *in principle,* to be one-valued, and we were either prevented from using them in connection with ∞-valued orders of abstractions, or, if used by semantic necessity, we identified the indefinitely many values into one. Both results were undesirable; the first established semantic blockages to creative scientific work; the other promulgated semantic disturbances. But once the multiordinality of terms is established, we have ∞-valued terms to which, in a given context (by differentiating the different orders of abstractions which a context indicates), we can ascribe single values.

Such a pioneering analysis may appear difficult at first, but this is only due to the lack of familiarity and established pre-A and A one-, two-, three-, or few-valued *s.r*, all of which involve ultimately *identification* somewhere. Once identification is abolished, however, and this is childishly simple, although not easy and rather laborious for grown-ups, ∞-valued semantics become natural and *automatic,* evading very serious theoretical difficulties. In the present volume, I had to elaborate in detail upon different issues, simply because my readers will be mostly grown-ups with established *pre-A*, and A reactions, who must first be made to recognize the benefits of a $\bar{A}$ evaluation before they will be willing to undergo a laborious re-education of their older *s.r*. The procedure in the training of infants and children is extremely simple and entirely on their levels.

There is, however, one point that I wish to make entirely clear. From the older point of view, one might say that a $\bar{A}$-system may lead to 'over-rationalization' and, consequently, take 'all the joy out of life'. Such objections are entirely unjustified. First of all, the A-system leads to shallow, but often clever verbal interplay of definitions, mostly non-similar in structure to the world and ourselves, representing a species of apologetics, and usually called 'rationalization'. The $\bar{A}$-system leads to

structural adjustment of language and *s.r*, and a *structural* enquiry, resulting in *understanding*. It makes shallow infantile 'rationalization', 'wishful thinking', and apologetics of different brands impossible, but leads to a higher order of adult intelligence, based on *proper evaluation*. In mere 'rationalization', we often have clever, but shallow, infantile evaluation, based on the ignorance or disregard of structural facts, which alone make up the content of all 'knowledge'. In a $\bar{A}$-system, by eliminating the sources of infantile evaluation and reactions, we supply the nervous system of the infant with uniquely appropriate material, so that it may develop into a 'normal' adult. In the older system, instead of helping, we hindered the development of adult standards of evaluation, with well-known results. There is nothing wrong with 'human nature' or the majority of nervous systems as such, but there is something definitely wrong with our educational methods inside and outside our schools.

There is another point which is still more convincing and, perhaps, even more decisive. The above-mentioned older objections are due to *s.r* based on the play upon *elementalistic* terms and are a *neurological impossibility*. The organism works as-a-whole, and in the cyclic nerve currents it is impossible, by any known educational methods, to abolish 'emotions'. But what *can be* accomplished is this: By training in silence on the un-speakable objective levels and in differentiation between different orders of abstractions, we automatically abolish the infantile identifications and evaluations; we introduce a 'delay in action', which is the physiological means for getting our 'emotions' under control and for engaging the fuller co-operation of the cortex. Infantile 'over-emotionalism' is abolished in the adult. Infants would behave as infants, but this infantile behaviour would not be carried into the period when adulthood should begin. The 'emotions' are not abolished but 'sublimated'.

It is true that many standards would be changed. For instance, we might roughly say that an infantile type is often bored by a symphony and that jazz satisfies his infantile make-up. If we were to take such an infantile grown-up and compel him to listen only to symphonies, this would not be kind, nor would it transform his infantile *s.r* into adult reactions. But, if unhampered by inappropriate semantic and so neurological training, he would be free to develop normally into an adult, and his own preference would be toward a symphony rather than toward primitive throbbings, his enjoyment, then, would not be diminished, but, perhaps, made fuller.

Similar analysis could be made of all human interests, with the result that the forcing of adult standards on infantile types would remain unkind; but the sad part of it is that, in spite of repressions, impositions . , these imposed standards remain largely ineffective and are abandoned as soon as compulsion ends. Not so, if, by proper semantic education, we allow the infant to develop normally into adulthood. The *new* standards are not imposed, but become his own. We do not then need compulsion from without, because the new standards act from within and become pleasurable and lasting.

A similar process is very obvious in the practice of psychotherapy. The standards of evaluation of the patients are usually inappropriate to the conditions of modern life and often clash sharply with the accepted standards. Moralizing without changing by *other means* his standards of evaluation never accomplishes any satisfactory therapeutic results; quite the contrary, it often does a great deal of harm. A physician would be very unwise to censure or condemn a symptom, as this would preclude any beneficial results. What physicians usually do is to treat any symptom, no matter how repulsive, with great sympathy and understanding. They *do not attempt* to change the symptom directly, but, by the understanding of its main mechanism, they try to *change the patient's standards of evaluation,* of which the symptom is only a consequence. If at all successful and the physician succeeds in changing the inappropriate standards of evaluation, the symptom then automatically disappears. In daily life, we usually attack only the symptoms, disregarding mostly the underlying structural foundations; this method accounts for the doubtful results.

Under infantile standards we apply similar methods to society. Many may want to abolish wars, revolutions, 'depressions'. , but they do not investigate structurally deep enough. They attack the symptoms, instead of analysing the structural issues which produce these symptoms.

In conclusion, let us notice that the analysis of a semantic mechanism on a printed page requires new terms and the co-ordination of many details . , which, at first, do not always appear so simple, although, once the theoretical side is mastered, the educational application is genuinely simple. Thus, the analysis of the one-, two-, three- . , and ∞-valued semantics may appear difficult, yet, in practice, it only amounts to imparting through our educational systems a semantic flexibility, instead of fixity; to acquiring the inclination of starting with observations, followed by descriptions, from which we pass to inferences, in connection with awareness of these ordered processes , . In training, it is enough to abolish identification, and this is easily achieved once we have produced

the proper method, based on a language of new $\bar{A}$ structure. This last actually consists of but a few, new, simple, and common-sense terms, the analysis of which helps us to discover a few simple and invariant psycho-physiological relations. Thus, identification is eliminated by starting with an *ordinal* language and method. Once we get the feel of the horizontal and vertical stratification, and learn to differentiate between orders of abstractions, identification disappears. Silence on the objective levels produces a 'delay', involves and trains the cortex; our reactions become more and more intelligent in the human sense., .; and the most important results are reached by the simplest means.

The training in visualization and the abolishing of objectification are the first and most important steps for a complete elimination of identification. When this first step is achieved, the rest is comparatively a very simple task.

But the reader may ask why we should have to use such unfamiliar and, therefore, seemingly difficult methods to achieve such obvious results. Do we really need a $\bar{A}$-system to achieve the results which, even in an A-system, are known to be desirable? The answer is weighty with consequences and should be taken very earnestly. In the A-system, these desirable results could not be attained generally, because the structure of our old languages and the method hampered rather than helped us. New theories, new systems., are built precisely with the aim to facilitate adjustment. Those questions which in the older days were supposed to be 'philosophical', 'metaphysical'., the application of which required a high grade of intelligence, knowledge., to start with, become, in the new way, simply *a problem of the structure of the language we use*. All the issues appear closely interrelated. We do not require 'high intelligence' nor 'higher education' to begin with, in order to obtain these desirable results, as the results follow *automatically* from the structure of the language we accept and teach our children. Thus, the older impossibilities are achieved simply and automatically, with the greatest possible efficiency and the most lasting results.